

A PROJECT REPORT ON

Virtual Agent–Driven SLA Breach Awareness & Justification System

Domain:service now system administrator

Submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING

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In Coordination with the Virtual Internship Program

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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CERTIFICATE OF COMPLETION

This is to certify that the project entitled "Virtual Agent–Driven SLA Breach Awareness & Justification System" has been carried out by

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of the Department of Computer Science and Engineering, SRM University AP, as a partial fulfillment of the requirements for the award of the Bachelor of Technology degree.

The work embodied in this report is original and has been carried out under the prescribed academic guidelines.

Internal Project Guide
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We also acknowledge ServiceNow for providing the Developer Instance, which enabled us to implement and validate our solution using industry-standard tools such as Flow Designer, Task SLA, Virtual Agent, Reporting, and Dashboards.

Finally, we extend our heartfelt thanks to our teammates and friends for their cooperation, constructive suggestions, and constant motivation during every phase of the project.

— Bhavani prasad kadiyala

ABSTRACT

Service Level Agreements (SLAs) are critical in IT Service Management (ITSM) as they ensure incidents and service requests are resolved within predefined timelines. In many organizations, SLA breaches occur because assigned users are unaware of approaching deadlines or fail to provide proper justification for delays. Existing systems often rely solely on email notifications, which may be overlooked or ignored.

The Virtual Agent–Driven SLA Breach Awareness & Justification System addresses this problem by integrating ServiceNow Virtual Agent, Flow Designer, Task SLA, UI Policies, Reports, and Dashboards into a unified SLA governance solution. The system automatically detects when an incident reaches a predefined SLA risk threshold and immediately notifies the assigned user through email and an interactive Virtual Agent conversation.

The Virtual Agent guides users through acknowledging the SLA risk and entering the appropriate breach reason and justification before the SLA is violated. Custom UI Policies ensure that only authorized assignees can provide justification, maintaining accountability and data integrity. Simultaneously, automated dashboards and reports provide managers with real-time visibility into SLA performance, pending justifications, acknowledged incidents, and breach trends.

The implementation demonstrates how intelligent automation can improve SLA compliance, reduce manual follow-up efforts, enhance communication between stakeholders, and support proactive incident management. Experimental validation confirms that the proposed system successfully automates SLA awareness, improves governance, and delivers actionable insights through reporting and analytics.

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CHAPTER 1

INTRODUCTION

1.1 Project Overview

In modern IT organizations, timely resolution of incidents is essential to maintain business continuity and customer satisfaction. Service Level Agreements (SLAs) define the maximum allowable time for resolving incidents based on business priorities. Failure to meet these agreements can lead to operational disruptions, financial losses, and reduced customer confidence.

Although ServiceNow provides robust SLA management capabilities, many organizations still experience SLA violations due to delayed user response, lack of awareness, inadequate communication, or missing documentation explaining the reason for delays. These challenges reduce transparency and complicate SLA governance.

The **Virtual Agent–Driven SLA Breach Awareness & Justification System** introduces an intelligent, automated approach for handling SLA risks. The solution continuously monitors incident SLAs using Task SLA records. When an SLA reaches the predefined risk threshold, Flow Designer automatically sends email notifications and invokes the ServiceNow Virtual Agent.

The Virtual Agent communicates directly with the assigned user, explaining that the SLA is approaching breach and requesting acknowledgement. Users are guided through entering the SLA breach reason and justification within the incident record. Custom UI Policies ensure that only authorized assignees can update these mandatory fields.

Managers and administrators can monitor SLA compliance using real-time dashboards and analytical reports, including pending justifications, breach reasons, acknowledged versus unacknowledged incidents, and overall SLA performance. This integrated workflow improves accountability, accelerates response times, and strengthens organizational SLA governance.

1.2 Purpose

The primary purpose of this project is to enhance SLA management by combining intelligent automation with interactive user engagement. Instead of relying solely on static notifications, the proposed system actively guides users through the actions required to prevent SLA breaches and document necessary justifications.

The project aims to:

- Reduce SLA violations through proactive notifications.
- Improve user awareness using ServiceNow Virtual Agent.
- Automate SLA monitoring using Task SLA.
- Capture standardized SLA breach reasons and justifications.
- Enhance transparency through dashboards and reports.
- Support managers in monitoring SLA compliance.
- Improve governance and accountability within IT Service Management.
- Minimize manual follow-up by automating communication workflows.
- Increase overall operational efficiency using ServiceNow automation capabilities.

The proposed system demonstrates how multiple ServiceNow components can be integrated to create a comprehensive SLA governance framework that improves incident management and organizational productivity.

CHAPTER 2

IDEATION PHASE

2.1 Problem Statement

Service Level Agreements (SLAs) are one of the most important components of IT Service Management (ITSM). They define the agreed time limits within which incidents and service requests must be resolved. Organizations rely on SLAs to ensure timely service delivery, maintain customer satisfaction, and measure the efficiency of support teams.

Despite the availability of SLA management features in ServiceNow, many organizations continue to experience SLA breaches. One of the primary reasons is that assigned users are often unaware that an SLA is approaching its deadline. In many cases, notifications sent through email are overlooked or ignored, resulting in delayed responses and unresolved incidents.

Another major challenge is the lack of accountability when SLA breaches occur. Most organizations require users to provide a valid reason and justification for exceeding SLA timelines. However, users frequently forget to enter this information or provide incomplete details, making it difficult for managers to analyze recurring issues and implement corrective actions.

Existing systems generally focus on notifying users rather than actively assisting them through the required actions. As a result, managers spend considerable time following up with assignees, collecting justifications manually, and preparing compliance reports.

To overcome these limitations, there is a need for an intelligent, automated solution that not only alerts users about SLA risks but also guides them through acknowledgement and justification while simultaneously providing management with real-time analytical dashboards and reports.

The proposed **Virtual Agent–Driven SLA Breach Awareness & Justification System** addresses these challenges by integrating ServiceNow Task SLA, Flow Designer, Virtual Agent, UI Policies, Reports, and Dashboards into a unified SLA governance framework.

Existing Problems

The existing SLA management process suffers from several limitations:

- Users often overlook email notifications regarding SLA deadlines.
- SLA breaches occur due to delayed acknowledgement.
- Manual collection of breach justifications consumes administrative effort.
- Lack of standardized breach reasons reduces reporting accuracy.
- Managers have limited visibility into pending SLA risks.
- Compliance monitoring requires multiple manual processes.
- Existing notification systems provide alerts but not guided assistance.
- Inconsistent documentation affects audit readiness.

Proposed Solution

The proposed solution introduces intelligent automation into the SLA lifecycle.

Whenever an SLA reaches the configured "At Risk" threshold:

1. Task SLA continuously monitors incident progress.
2. Flow Designer detects SLA risk.
3. Automated email notification is generated.
4. Virtual Agent immediately engages the assigned user.
5. User acknowledges the SLA risk.
6. UI Policy enables mandatory justification fields.
7. User submits SLA breach reason and justification.
8. Dashboard and reports update automatically.
9. Managers monitor SLA compliance in real time.

This proactive workflow minimizes SLA breaches and improves organizational accountability.

2.2 Empathy Map Canvas

The Empathy Map helps understand the needs and behavior of different stakeholders involved in the SLA management process.

9. EMPATHY MAP



Primary User

Incident Assignee / Support Engineer

Says

- "I was not aware the SLA was about to expire."
- "I forgot to update the justification."
- "I need reminders before the SLA breaches."
- "I need an easier way to complete compliance tasks."

Thinks

- "There are too many incidents to monitor."
- "Emails are easy to miss."
- "I hope I don't miss the SLA deadline."
- "The system should guide me automatically."

Does

- Works on assigned incidents.
- Updates incident progress.
- Responds to Virtual Agent notifications.
- Provides SLA breach justification.
- Resolves incidents.

Feels

- Pressure to meet SLA deadlines.
- Concern about managerial review.
- Confusion when multiple incidents become overdue.
- Relief when guided through the required process.

Secondary User

SLA Manager

Says

- "I need better SLA visibility."
- "Why was this SLA breached?"
- "Which users haven't acknowledged SLA risks?"

Thinks

- "Can we reduce SLA violations?"
- "Which teams require additional training?"
- "How can we automate compliance?"

Does

- Reviews dashboards.
- Monitors reports.
- Approves governance policies.
- Evaluates SLA trends.

Feels

- Responsible for compliance.
- Needs accurate analytics.
- Wants proactive monitoring.

2.3 Brainstorming Initial Ideas Considered

During the ideation phase, several approaches were explored to improve SLA governance.

Idea 1

Only send reminder emails.

Limitation:

Users may ignore or overlook emails.

Idea 2

Escalate incidents directly to managers.

Limitation:

Creates unnecessary escalations without involving the assignee first.

Idea 3

Create manual SLA review meetings.

Limitation:

Time-consuming and not scalable.

Idea 4

Develop automated dashboards only.

Limitation:

Dashboards identify problems but do not prevent SLA breaches.

Idea 5 (Selected Solution)

Integrate:

- Task SLA
- Flow Designer
- Email Notifications
- Virtual Agent
- UI Policies
- Reports

- Dashboards

This solution provides proactive notifications, guided user interaction, automated justification collection, and real-time managerial visibility.

Brainstorming Outcome

The team concluded that combining multiple ServiceNow capabilities into a single workflow would provide the most effective SLA governance solution.

The selected architecture enables:

- Early detection of SLA risks.
- Immediate communication with assignees.
- Interactive guidance using Virtual Agent.
- Automated justification capture.
- Real-time dashboard updates.
- Improved managerial decision-making.
- Reduced manual effort.
- Increased SLA compliance.

Benefits Identified During Brainstorming

- Reduction in SLA breaches.
 - Improved employee awareness.
 - Faster incident resolution.
 - Better documentation for audits.
 - Higher customer satisfaction.
 - Reduced administrative workload.
 - Enhanced reporting capabilities.
 - Improved ITSM governance.
 - Greater accountability across support teams.
-

CHAPTER 3

REQUIREMENT ANALYSIS

Requirement analysis is one of the most important phases in software development because it identifies the functional and technical requirements needed to develop a successful system. In this phase, the requirements of users, managers, and administrators were studied to design an automated SLA governance solution. The proposed system was developed after analyzing the workflow of incident management in ServiceNow and identifying areas where automation could improve SLA compliance.

The primary objective of the requirement analysis phase was to understand the interaction between incident assignees, SLA managers, and the ServiceNow platform while ensuring efficient monitoring of Service Level Agreements (SLAs).

3.1 Customer Journey Map

The Customer Journey Map illustrates how different users interact with the proposed system from incident creation to SLA monitoring and justification submission.

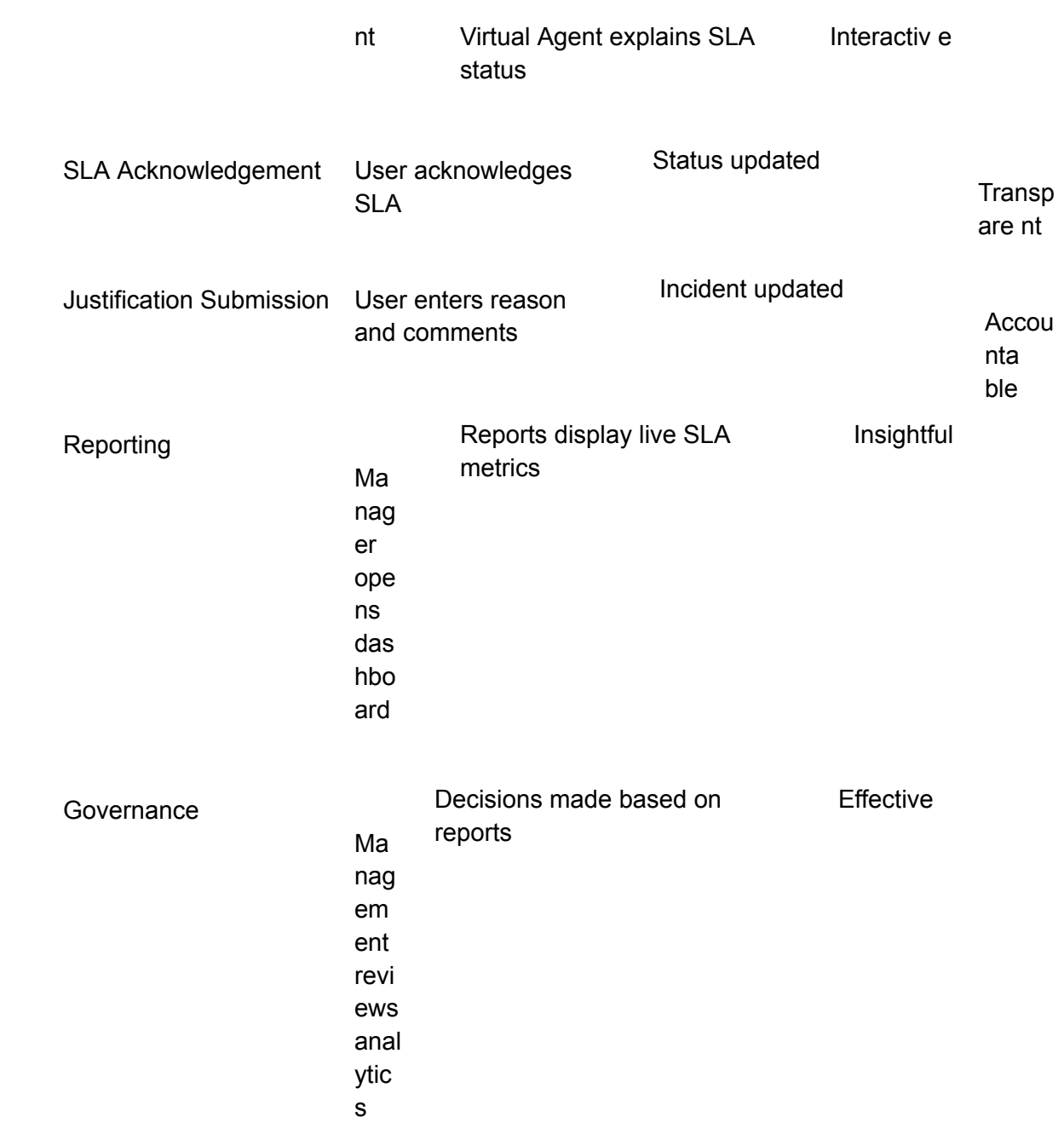
8. CUSTOMER JOURNEY MAP

Stage	User Action	System Response	User Feeling
 Incident Creation	Create Incident	Incident is recorded and SLA attached	Satisfied
 SLA Monitoring	Works on Incident	SLA is running and monitored	Normal
 SLA At Risk	Receives Email Alert	Email notification is sent	Concerned
 Virtual Agent	Chats with Virtual Agent	Guides and requests acknowledgement	Engaged
 Acknowledge	Acknowledge SLA Risk	Acknowledgement recorded	Responsible
 Justification	Enters Reason & Justification	Information saved in incident	Complete
 Monitoring	Manager views Dashboard	Dashboard & reports are updated	Confident

Stage	User Action	System Response	User Experience
Incident Creation	User create	Incident is recorded in ServiceNow	

Smooth

Incident Assignment	Incident is assigned to support engineer	Task SLA starts automatically	Efficient
SLA Monitoring	System continuously monitors SLA	Business time percentage updates	Automatic
SLA At Risk	SLA reaches configured threshold	Flow Designer triggers notification	User alerted
Email Notification	User receives warning email	Email contains Virtual Agent guidance	Informative



Customer Journey Description

The journey begins when an incident is created within the ServiceNow Incident Management module. Once assigned, the Task SLA engine automatically starts tracking the SLA timer.

As the incident approaches the predefined SLA threshold, Flow Designer detects the risk and sends an automated email notification to the assigned engineer. The email instructs the user to acknowledge the SLA risk through the Virtual Agent.

The Virtual Agent interacts with the user by explaining the SLA status and requesting

acknowledgement. If necessary, the user provides a breach reason and justification through controlled UI Policy fields.

Finally, managers monitor SLA compliance using reports and dashboards, enabling proactive governance and better decision-making.

3.2 Solution Requirement

The proposed system consists of both functional and non-functional requirements.

Functional Requirements

Requirement ID	Description
FR-01	Create Incident Records
FR-02	Automatically attach Task SLA
FR-03	Monitor SLA Business Percentage
FR-04	Detect SLA At Risk condition
FR-05	Send automated email notifications
FR-06	Launch Virtual Agent conversation
FR-07	Allow SLA acknowledgement
FR-08	Capture breach reason
FR-09	Capture detailed justification
FR-10	Apply UI Policies for authorized users
FR-11	Generate SLA Reports
FR-12	Display SLA Dashboard

Explanation of Functional Requirements

Incident Creation

The system shall allow users to create incidents with relevant details such as caller, short description, priority, and assignment group.

Task SLA Monitoring

Whenever an incident is created, the corresponding Task SLA shall automatically attach and begin tracking business time.

SLA Risk Detection

The system shall continuously monitor the SLA percentage and identify incidents approaching breach.

Automated Notifications

Flow Designer shall automatically notify users through email whenever an SLA reaches the configured risk threshold.

Virtual Agent Guidance

The Virtual Agent shall provide an interactive conversation guiding users through acknowledgement and justification.

SLA Justification

Only the assigned engineer shall be permitted to provide breach reasons and justifications using controlled UI Policies.

Reporting

The system shall generate reports including:

- SLA Breaches by Reason
- Pending Justifications

- Acknowledged vs Unacknowledged Incidents
- SLA Governance Dashboard

Non-Functional Requirements

Requirement	Description
Availability	System available 24×7
Reliability	Accurate SLA monitoring
Performance	Fast response time
Scalability	Supports increasing incidents
Security	Role-based access
Maintainability	Easy configuration
Usability	User-friendly Virtual Agent
Compatibility	Compatible with ServiceNow modules

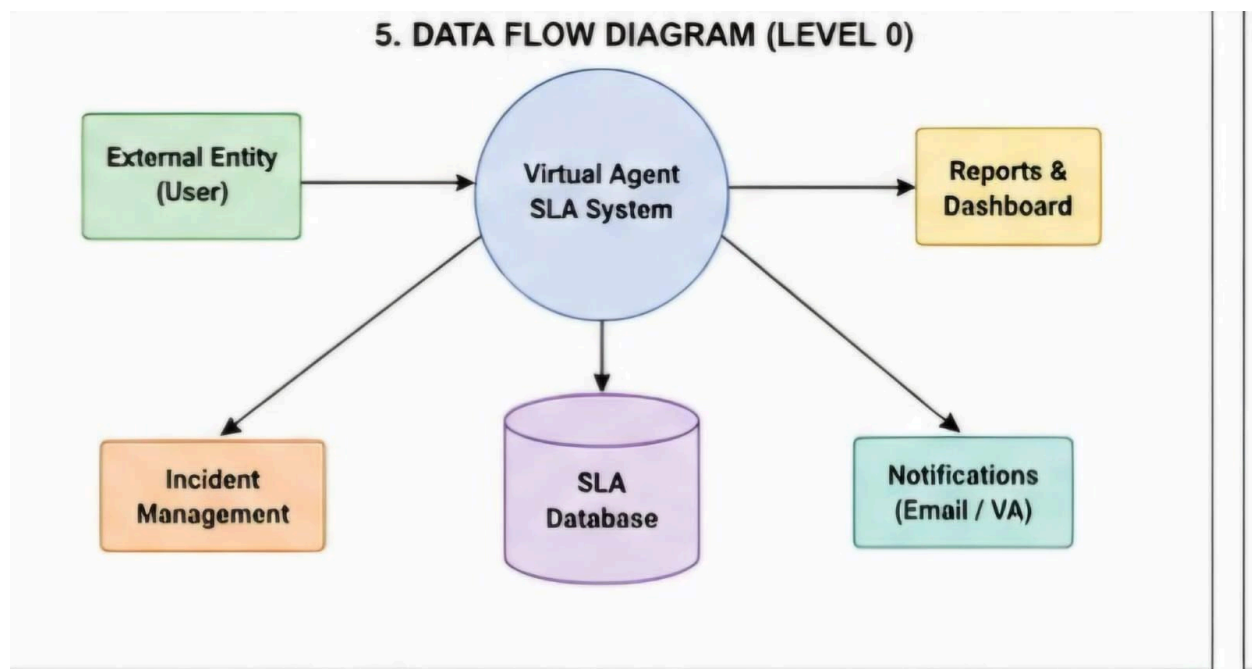
Performance Requirements

The proposed solution should satisfy the following performance criteria.

Parameter	Expected Performance
Incident Creation	Less than 2 seconds
SLA Attachment	Immediate
Email Notification	Within 1 minute
Virtual Agent Response	Less than 2 seconds
Dashboard Loading	Less than 5 seconds
Report Generation	Less than 10 seconds

3.3 Data Flow Diagram

Data Flow Diagram (Level 0)



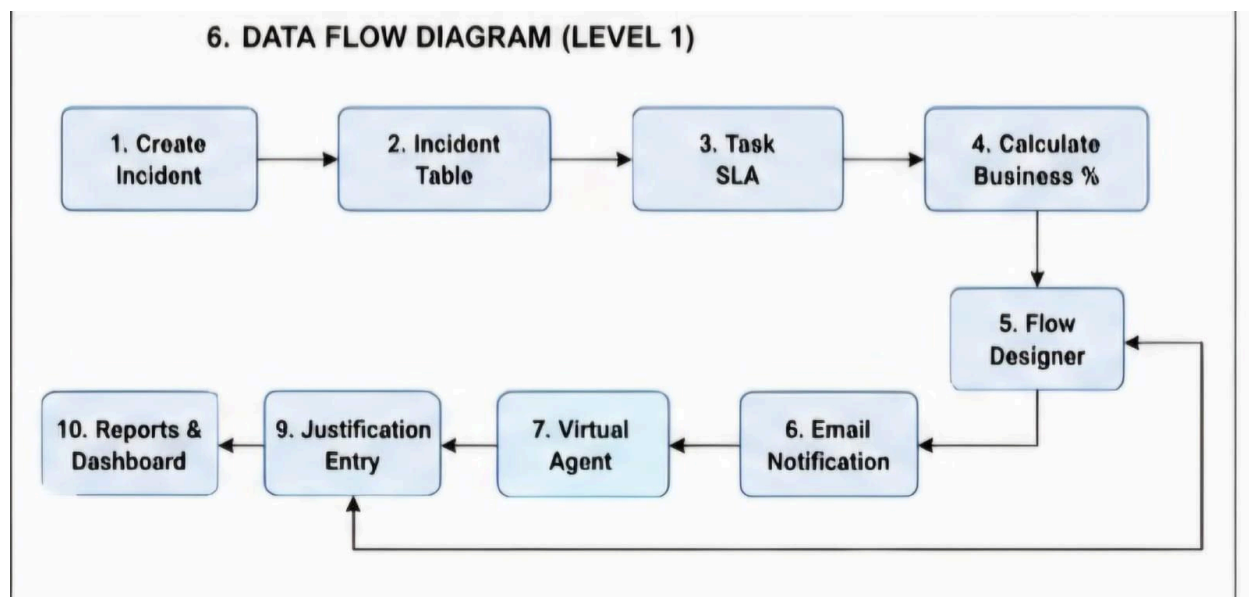
Description

The Level 0 Data Flow Diagram represents the overall interaction between the user and the proposed SLA governance system.

Explanation

The user initiates the workflow by creating an incident. ServiceNow automatically attaches the appropriate Task SLA and begins monitoring the SLA timer. When the SLA approaches its configured threshold, Flow Designer sends notifications through email and triggers the Virtual Agent. The user acknowledges the SLA risk, submits the required justification, and the system updates reports and dashboards for managerial review.

Data Flow Diagram (Level 1)



DFD Explanation

The Level 1 Data Flow Diagram explains the internal workflow of the proposed system. The Task SLA continuously calculates the business percentage. If the SLA exceeds the predefined threshold, Flow Designer automatically initiates email notifications and Virtual Agent interactions. Users then acknowledge the SLA, provide justifications, and the information is stored for reporting and dashboard visualization.

3.4 Technology Stack

The proposed solution is implemented entirely using the ServiceNow platform and its built-in capabilities.

Technology	Purpose
ServiceNow	Core development platform
Incident Management	Incident handling
Task SLA	SLA monitoring
Flow Designer	Workflow automation
Virtual Agent	User interaction
UI Policies	Dynamic field control
Business Rules	Backend logic
Email Notifications	Automated alerts
Reports	SLA analytics
Dashboards	Governance monitoring
JavaScript	Server-side scripting
Glide API	ServiceNow API integration

Software Requirements

Software	Version
ServiceNow Developer Instance	Zurich Release
Web Browser	Google Chrome / Microsoft Edge
Operating System	Windows 10 / Windows 11
Visual Studio Code	Optional
Microsoft Word	Documentation

Hardware Requirements

Component

Specification

Processor	Intel Core i3 or
above RAM	8 GB or above
Storage	20 GB Free Space
Internet	Stable Broadband Connection

Advantages of Selected Technology Stack

- Cloud-based implementation.
- No additional infrastructure required.
- Low maintenance cost.
- Built-in security mechanisms.
- Rapid application development.
- Easy integration with ITSM modules.
- Real-time reporting capabilities.
- Scalable architecture suitable for enterprise deployment.

CHAPTER 4

PROJECT DESIGN

The Project Design phase converts the identified requirements into a practical system architecture that can be implemented in ServiceNow. This phase defines how different ServiceNow modules interact to automate SLA monitoring, notify users, collect justifications, and generate management reports.

The proposed design follows a modular approach in which each ServiceNow component performs a specific responsibility while remaining integrated with the overall workflow.

4.1 Problem–Solution Fit

Existing Problem

Organizations using Incident Management often encounter SLA breaches because assigned engineers are unaware of approaching deadlines or fail to provide timely justifications. Although ServiceNow supports SLA tracking, it does not proactively guide users through the acknowledgement and documentation process.

The existing process faces several issues:

- Email notifications are frequently ignored.
- Engineers are unaware of SLA status.
- Manual follow-up by managers is required.
- Justifications are often incomplete or missing.
- SLA reports lack standardized breach reasons.
- No interactive mechanism exists to engage users before a breach occurs.

These issues reduce operational efficiency and increase SLA violations.

Proposed Solution

The proposed solution integrates multiple ServiceNow capabilities into a single automated workflow.

When an incident reaches a predefined SLA threshold:

- Task SLA continuously monitors business elapsed time.
- Flow Designer detects the SLA risk.

- Email notifications are automatically sent.
- Virtual Agent immediately interacts with the assigned engineer.
- Users acknowledge the SLA risk.
- UI Policies enable mandatory justification fields.
- Reports and dashboards are updated automatically.

This design ensures that users are informed, guided, and accountable throughout the SLA lifecycle.

Problem–Solution Mapping

Existing Problem	Proposed Solution
Users ignore SLA emails	Interactive Virtual Agent notifications
Manual follow-up	Automated Flow Designer workflows
Missing justification	Mandatory UI Policy fields
Poor visibility	SLA Dashboard
Delayed action	Early SLA At Risk detection
Manual reporting	Automated Reports
Lack of governance	Centralized SLA Dashboard

Why This Solution Was Selected

The team selected this architecture because it provides:

- Complete automation
- Minimal manual intervention
- Improved user engagement
- Better SLA compliance
- Real-time managerial visibility
- Enterprise scalability
- Seamless integration with existing ServiceNow ITSM modules

4.2 Proposed Solution

The proposed solution consists of several integrated modules working together.

Module 1 – Incident Management

Users create incidents through the ServiceNow Incident application.

Information collected includes:

- Caller
- Category
- Assignment Group
- Priority
- Short Description
- Description

After creation, the incident is assigned to the appropriate support engineer.

Module 2 – Task SLA Engine

Whenever an incident is assigned, Task SLA automatically:

- Attaches the SLA
- Calculates business time
- Calculates elapsed percentage
- Monitors SLA status
- Detects SLA risk

No manual intervention is required.

Module 3 – Flow Designer

Flow Designer continuously evaluates SLA conditions.

When:

Business Percentage $\geq$ Configured Threshold

Flow Designer automatically performs:

- Email Notification
- Virtual Agent Trigger
- Incident Update
- Dashboard Refresh

This module serves as the automation engine.

Module 4 – Email Notification

When the SLA reaches the configured threshold:

The assigned engineer receives an email containing:

- Incident Number
- Current SLA Status
- Warning Message
- Virtual Agent Link
- Dashboard Link

This ensures users are informed immediately.

Module 5 – Virtual Agent

The Virtual Agent provides an interactive conversational interface.

Functions include:

- Informing users about SLA risk
- Requesting acknowledgement
- Guiding users through justification
- Answering common SLA questions
- Redirecting users to the incident

This significantly improves user engagement compared to traditional email alerts.

Module 6 – UI Policy

UI Policies dynamically control incident fields.

When the assigned engineer opens the incident:

- Justification field becomes editable.
- Breach reason becomes mandatory.
- Other users cannot modify protected fields.

Module 7 – Reports

Reports provide analytical insights including:

- SLA Breaches by Reason
- Pending Justifications
- Acknowledged Incidents
- Unacknowledged Incidents
- Monthly SLA Trends

Managers use these reports for operational analysis.

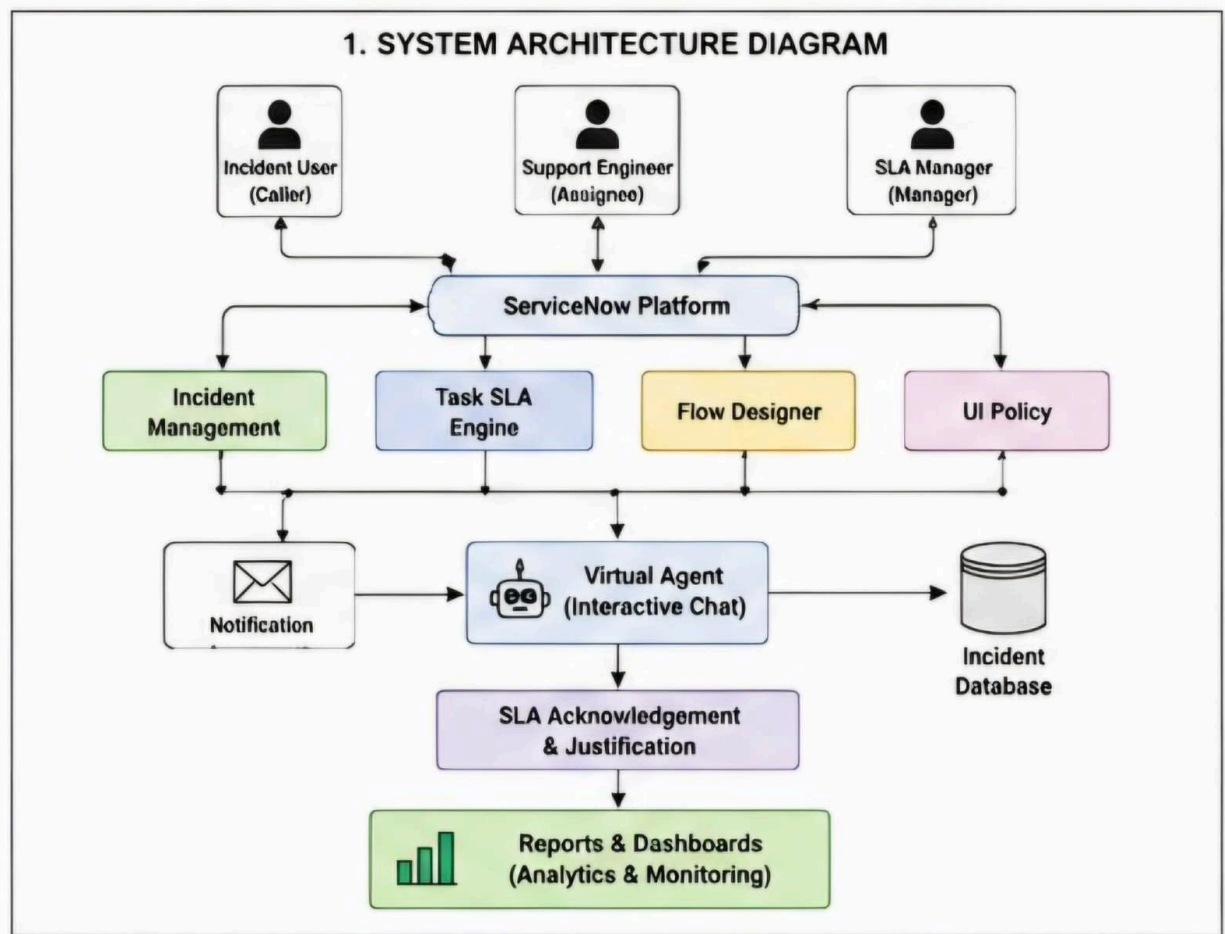
Module 8 – Dashboard

The SLA Governance Dashboard displays:

- Total Incidents
- Active SLAs
- At Risk Incidents
- Breached SLAs
- Pending Justifications
- SLA Compliance Percentage

The dashboard provides a real-time overview of organizational SLA performance.

4.3 Solution Architecture



The architecture of the proposed system consists of six primary layers.

Layer 1 – User Layer

Users include:

- Incident Caller
- Support Engineer
- Team Lead
- SLA Manager
- Administrator

These users interact with the ServiceNow platform through a web browser.

Layer 2 – Application Layer

This layer contains:

- Incident Module
- Task SLA
- Virtual Agent
- Flow Designer
- UI Policies

These components perform business operations.

Layer 3 – Business Logic Layer

Business Rules manage:

- SLA calculations
- Incident validation
- Field updates
- Access control
- Automation logic

Layer 4 – Notification Layer

Responsible for:

- Email Notifications
- Virtual Agent Conversations
- SLA Alerts

This layer ensures proactive communication.

Layer 5 – Analytics Layer

Contains:

- Reports
- Dashboards
- SLA Statistics
- Performance Metrics

Used by managers for governance.

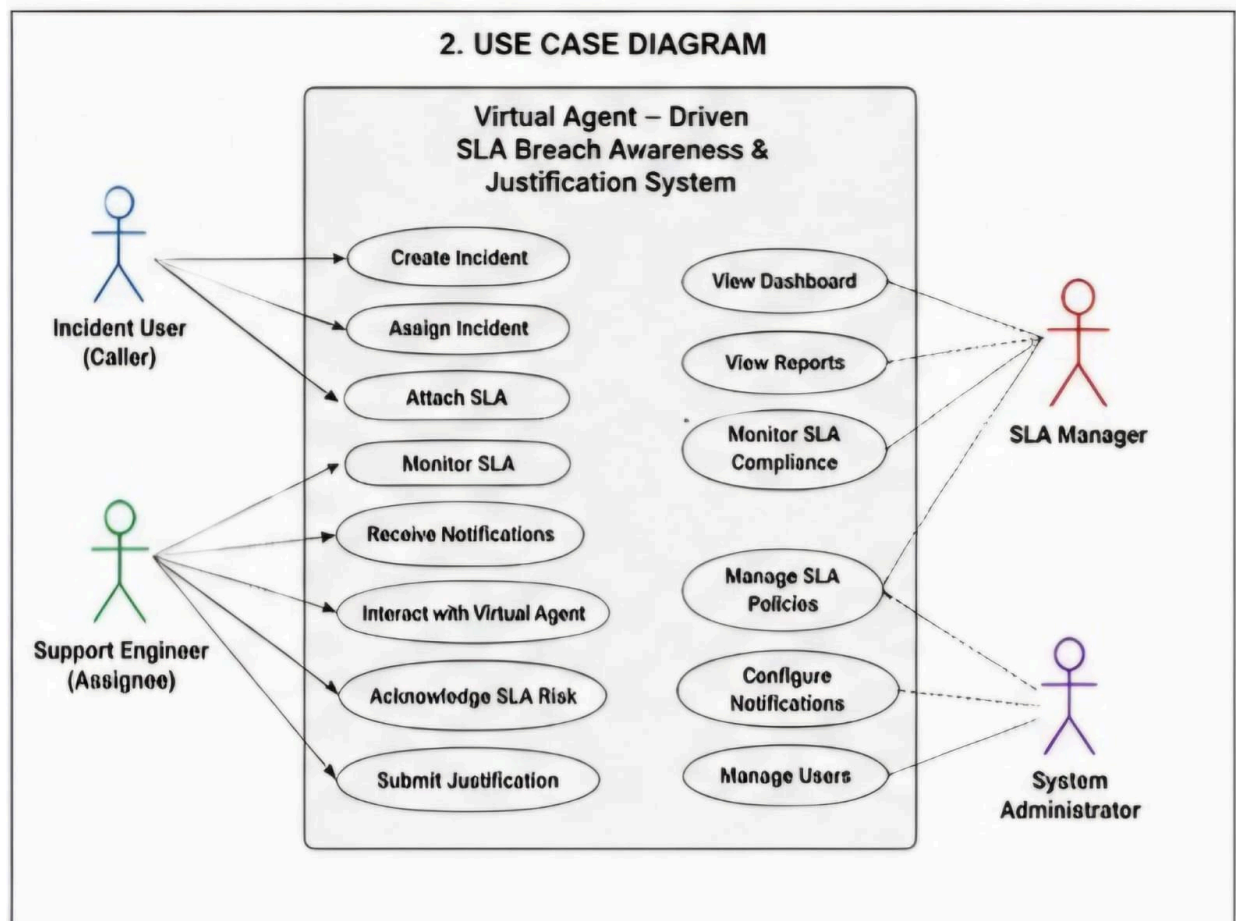
Layer 6 – Database Layer

ServiceNow stores:

- Incident Records
- Task SLA Records
- User Information
- Justifications
- Reports
- Dashboard Data

This provides centralized and secure data management.

Use Case Diagram



Actors

- Incident User

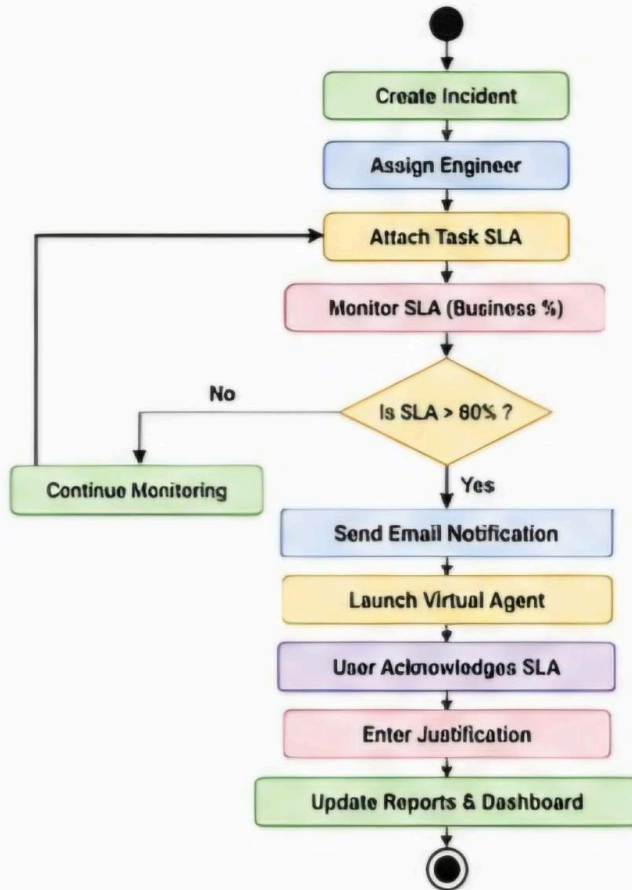
- Support Engineer
- SLA Manager
- System Administrator

Use Cases

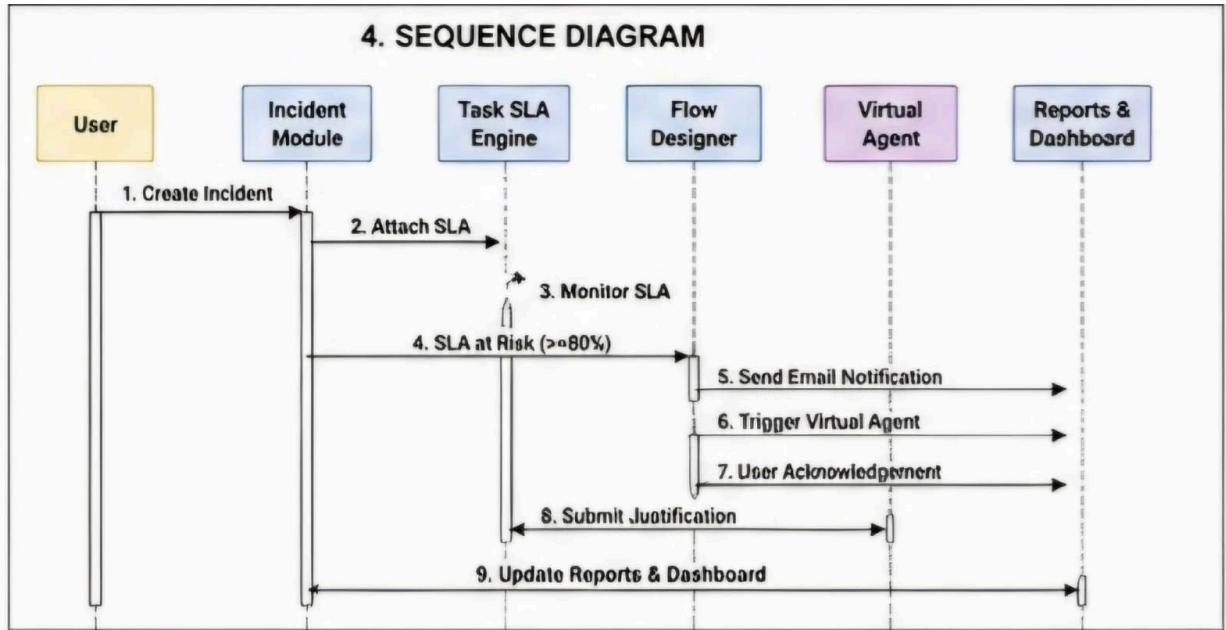
- Create Incident
- Assign Incident
- Attach SLA
- Monitor SLA
- Send Email Notification
- Launch Virtual Agent
- Acknowledge SLA
- Submit Justification
- Generate Reports
- View Dashboard
- Manage SLA Policies

Activity Diagram

3. ACTIVITY DIAGRAM



Sequence Diagram



Security Design

The proposed system incorporates the following security mechanisms:

- Role-based access control (RBAC)
- UI Policies for field-level permissions
- ServiceNow authentication
- Audit history tracking
- Secure data storage
- Controlled modification of justification fields
- Administrator privileges for configuration

Scalability

The architecture is designed to support:

- Multiple departments
- Large numbers of incidents
- Additional SLA definitions
- Future chatbot enhancements
- Integration with external ITSM tools

CHAPTER 5

PROJECT PLANNING & SCHEDULING

Project planning is an essential phase of software development that ensures systematic execution of project activities within the allocated time and resources. It involves identifying project objectives, dividing the work into manageable tasks, allocating responsibilities among team members, estimating timelines, identifying potential risks, and monitoring project progress. Effective planning helps ensure that the project is completed successfully while maintaining quality standards.

The **Virtual Agent–Driven SLA Breach Awareness & Justification System** was developed using a structured software development approach. The project was divided into several phases, including requirement analysis, system design, implementation, testing, deployment, and documentation. Each phase was planned carefully to ensure smooth coordination among team members and timely completion.

5.1 Project Planning

The project was executed following a sequential development approach where each phase was completed before moving to the next. The planning process focused on understanding the problem, designing the proposed solution, implementing the required ServiceNow components, testing the functionality, and preparing the final documentation.

Project Objectives

The major objectives identified during the planning phase were:

- Develop an automated SLA governance system.
- Monitor incident SLAs using Task SLA.
- Trigger automated notifications before SLA breaches.
- Integrate Virtual Agent for SLA acknowledgement.
- Collect SLA breach reasons and justifications.
- Generate analytical reports and dashboards.
- Improve SLA compliance through automation.
- Reduce manual intervention in SLA monitoring.

Project Development Methodology

The project followed the **Software Development Life Cycle (SDLC)** methodology consisting of six major phases.

Phase 1 – Requirement Gathering

The team analyzed the existing SLA management process and identified limitations such as delayed notifications, lack of user awareness, and manual justification tracking. User requirements and business objectives were documented to define the scope of the project.

Phase 2 – System Design

Based on the collected requirements, the architecture of the proposed system was designed. This included identifying the ServiceNow modules to be used, designing workflows, and preparing diagrams such as the architecture diagram, data flow diagrams, and use case diagram.

Phase 3 – Implementation

The implementation phase involved configuring the ServiceNow platform by creating custom fields, configuring Task SLAs, designing Flow Designer workflows, implementing UI Policies, creating Virtual Agent topics, and developing reports and dashboards.

Phase 4 – Testing

The implemented solution was validated through functional and performance testing. Eleven test cases were executed to verify that each module performed as expected. Testing included incident creation, SLA attachment, notification generation, Virtual Agent interaction, justification submission, and dashboard validation.

Phase 5 – Deployment

After successful testing, the project was deployed within the ServiceNow Developer Instance. The solution was verified using real-time incidents and SLA monitoring scenarios.

Phase 6 – Documentation

The final phase involved documenting the project, including architecture, implementation details, testing results, screenshots, and future enhancements.

Work Breakdown Structure (WBS)

The Work Breakdown Structure divides the project into smaller manageable tasks.

WBS ID	Activity	Description
1	Requirement Analysis	Study existing system and identify requirements
2	Literature Study	Understand ServiceNow modules and SLA concepts
3	System Design	Design architecture and workflow
4	Incident Configuration	Configure Incident Management
5	Task SLA Configuration	Create and configure SLA definitions
6	Flow Designer	Develop automation workflows
7	Email Notifications	Configure automated email alerts
8	Virtual Agent	Develop SLA awareness chatbot
9	UI Policy	Configure dynamic form behavior
10	Reports	Create SLA analytical reports
11	Dashboard	Design SLA Governance Dashboard
12	Functional Testing	Execute project test cases
13	Documentation	Prepare final report

Team Responsibilities

The project was developed collaboratively by four team members. Each member contributed to specific modules while coordinating with the team to ensure successful integration.

Team Member	Responsibility
Bhavani Prasad Kadiyala	Project coordination, Task SLA configuration, Flow Designer automation, report documentation
Venkata Chaitanya Adusumalli	Incident Management configuration, UI Policies, Business Rules
Pavalika Ganesh	Virtual Agent development, email notification templates, chatbot testing
Gurucharan Kalla	Reports, dashboards, testing, validation, documentation support

Risk Analysis

Several potential risks were identified during project planning.

Risk	Impact	Mitigation Strategy
Incorrect SLA configuration	High	Validate Task SLA settings before testing
Flow execution failure	Medium	Verify Flow Designer conditions and logs
Email notification delay	Medium	Test notification triggers with sample incidents
Virtual Agent errors	Medium	Validate conversation flow and variables
Dashboard data inconsistency	Medium	Verify report filters and data sources
User permission issues	High	Implement role-based access control and UI Policies

Resource Planning

The following resources were utilized for project development.

Software Resources

- ServiceNow Developer Instance (Zurich Release)
- Google Chrome / Microsoft Edge
- Microsoft Word
- Visual Studio Code (optional)

Hardware Resources

- Intel Core i3 Processor or above
- 8 GB RAM
- Stable Internet Connection
- Windows 10/11 Operating System

Deliverables

The project produced the following deliverables:

- Configured Incident Management Module
- Task SLA Configuration
- Flow Designer Automation
- Email Notification System
- Virtual Agent Topic
- UI Policy Configuration
- SLA Governance Dashboard
- Analytical Reports
- Functional Testing Report
- Final Project Documentation

Project Success Criteria

The project was considered successful if the following objectives were achieved:

- SLA monitoring performed automatically.
- Email notifications were generated successfully.
- Virtual Agent guided users through acknowledgement.
- Justification fields were enforced using UI Policies.
- Reports displayed accurate SLA information.
- Dashboard provided real-time governance metrics.
- All functional test cases passed successfully.

CHAPTER 6

FUNCTIONAL AND PERFORMANCE TESTING

Testing is an important phase in the Software Development Life Cycle (SDLC) that verifies whether the developed system satisfies all functional and non-functional requirements. The objective of testing is to identify defects, validate the implemented functionalities, and ensure that the system performs as expected under different operational conditions.

The **Virtual Agent–Driven SLA Breach Awareness & Justification System** was tested in the ServiceNow Developer Instance using multiple real-time scenarios. Each module, including Incident Management, Task SLA, Flow Designer, Email Notifications, Virtual Agent, UI Policies, Reports, and Dashboards, was validated independently as well as in an integrated environment.

The testing process confirmed that the developed solution successfully automates SLA monitoring, alerts users before SLA breaches, captures justifications, and provides managers with real-time governance reports.

6.1 Testing Methodology

The project followed **Black Box Testing** to validate the functionality of each module without considering its internal implementation.

The testing process included:

- Unit Testing
- Integration Testing
- Functional Testing
- Performance Testing
- User Acceptance Testing (UAT)

Each component was tested individually before validating the complete workflow.

6.2 Functional Testing

Functional testing verifies whether every feature works according to the specified requirements.

The following eleven test cases were executed successfully.

Test Case 1 – Incident Creation

Parameter

Description Test Case ID TC-01

Module	Incident Management
Objective	Verify incident creation
Precondition	User logged into ServiceNow
Test Steps	1. Open Incident Module 2. Create a new Incident 3. Enter mandatory details 4. Save the Incident

Expected Result

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Actual Result Incident created successfully

Status  PASS

ServiceNow (934) ServiceNow Backend C SkillWallet Emails View INC0010010 Operation: SLA at Risk Operation: ServiceNow ServiceNow

https://dev320543.service-now.com/incident.do?sys_id=d3e2bf1dc38a8f10e0d95fedd4013129&sysparm_record_target=incident&sysparm_record_row=1&sysparm_record_rows=42&sysparm_record_list...

Incident INC0010010

Follow Update Resolve Delete

Number INC0010010

* Caller System Administrator

Category Inquiry / Help

Subcategory -- None --

Service

Service offering

Configuration item

* Short description Test SLA Incident

Description

Channel -- None --

State In Progress

Impact 3 - Low

Urgency 3 - Low

Priority 5 - Planning

Assignment group SLA Governance Team

Assigned to John Assignee

Related Search Results

SLA At Risk

Acknowledgement Time

Notes

Related Records

Resolution Information

Watch list

Work notes list

Test Case 2 – SLA Attachment & Running

Parameter	Description	
Test Case ID	TC-02	
Module	Task SLA	
Objective	Verify automatic SLA attachment	
Precondition	Incident exists	
Test Steps	1. Open Incident 2. View Task SLA Related List	
Expected Result	SLA automatically attached and business percentage starts increasing	
Actual Result	SLA attached successfully	
Status	PASS	

https://dev320543.service-now.com/incident.do?sys_id=d3e2bf1dc38a8f10e0d95fedd4013129&sysparm_record_target=incident&sysparm_record_row=1&sysparm_record_rows=42&sysparm_record_lis...

Incident
INC0010010

System Administrator

Task SLA Incident Resolution - Virtual Agent Gove removed from 'task' table

2026-07-12 08:24:42

System Administrator

Field changes

2026-07-12 07:13:46

Assigned to

John Assignee

Impact

3 - Low

Incident state

In Progress

Opened by

System Administrator

Priority

5 - Planning

Update

Resolve

Delete

Related Links

[View SLA Timeline](#)

[Repair SLAs](#)

Task SLAs (1)

Affected CIs

Impacted Services/CIs

Child Incidents

SLA definition

Search

Actions on selected rows...

Task = INC0010010

SLA definition	Type	Target	Stage	Business time left	Business elapsed time	Business elapsed percentage	Start time	Stop time
Incident Resolution - Virtual Agent Gove	SLA	Resolution	In progress	6 Hours 37 Minutes	1 Hour 22 Minutes	17.23%	2026-07-12 07:13:46	(empty)

1 to 1 of 1

Test Case 3 – SLA At Risk Trigger

Parameter	Description
Test Case ID	TC-03
Module	Task SLA
Objective	Verify SLA At Risk trigger
Precondition	SLA reaches configured threshold
Test Steps	Wait until SLA reaches configured business percentage
Expected Result	SLA At Risk field becomes active
Actual Result	SLA At Risk successfully triggered
Status	✓ PASS

https://dev320543.service-now.com/incident.do?sys_id=d3e2bf1dc38a8f10e0d95fedd4013129&sysparm_record_target=incident&sysparm_record_row=1&sysparm_record_rows=42&sysparm_record_lis...

Incident INCO010010

System Administrator • 2026-07-12 08:24:42
Task SLA Incident Resolution - Virtual Agent Gove removed from 'task' table

System Administrator • Field changes • 2026-07-12 07:13:46

Assigned to: John Assignee
Impact: 3 - Low
Incident state: In Progress
Opened by: System Administrator
Priority: 5 - Planning

Update Resolve Delete

Related Links
[Show SLA Timeline](#)
[Repair SLAs](#)

Task SLAs (1) Affected CIs Impacted Services/CIs Child Incidents

SLA definition Search

Task = INCO010010

SLA definition	Type	Target	Stage	Business time left	Business elapsed time	Business elapsed percentage	Start time	Stop time
Incident Resolution - Virtual Agent Gove	SLA	Resolution	In progress	6 Hours 37 Minutes	1 Hour 22 Minutes	17.23%	2026-07-12 07:13:46	(empty)

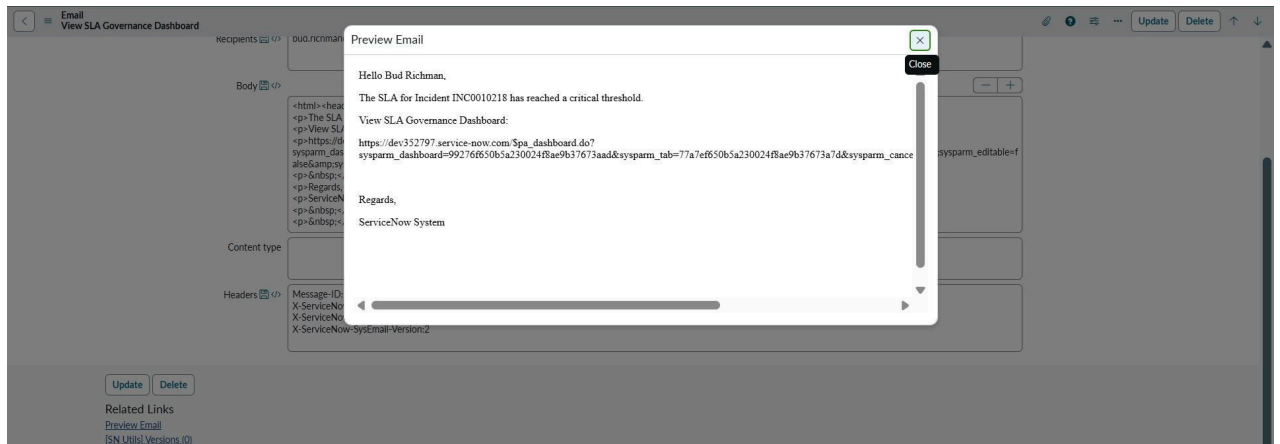
1 to 1 of 1

Test Case 4 – SLA Notification Email

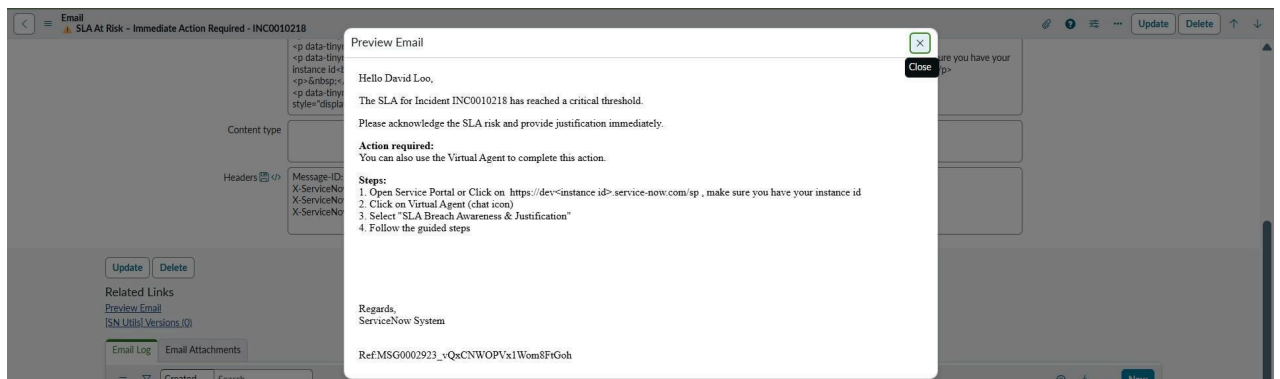
Parameter	Description
Test Case ID	TC-04
Module	Email Notification
Objective	Verify email notification
Precondition	SLA reaches threshold
Test Steps	Wait for Flow Designer to trigger notification
Expected Result	Email notification delivered successfully
Actual Result	Email received successfully
Status	 PASS

Screenshots:

- Figure 7.4(a)



- Figure 7.4(b)



Test Case 5 – UI Policy (Assignee Access)

Parameter	Description	
Test Case ID	TC-05	
Module	UI Policy	
Objective	Verify field-level control	
Precondition	Incident marked SLA At Risk	
Test Steps	Open Incident as assigned engineer	
Expected Result	Breach Reason and Justification fields become editable and mandatory	
Actual Result	UI Policy executed successfully	
Status	PASS	

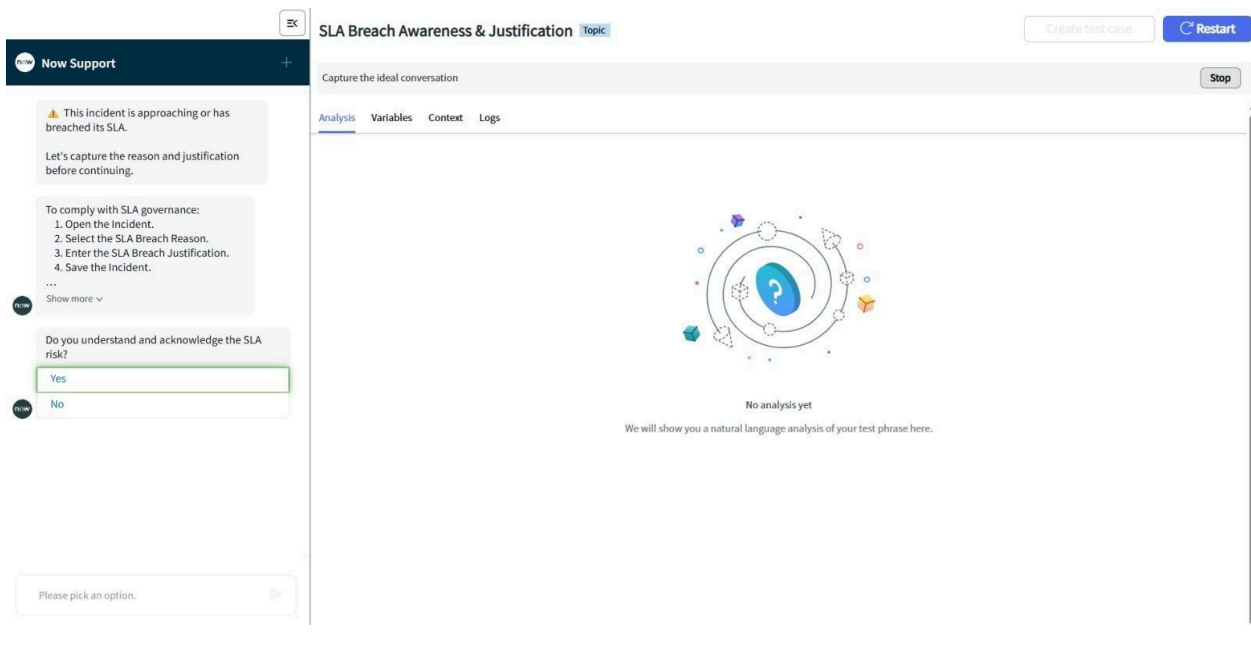
Screenshot: Figure 7.5 –

The screenshot shows a ServiceNow incident record form for incident INC0010010. The form is divided into two main sections. The top section contains fields for incident details: Number (INC0010010), Caller (System Administrator), Category (Inquiry / Help), Subcategory (-- None --), Service, Service offering, Configuration item, Short description (Test SLA Incident), and Description. The right side of the top section contains dropdown menus for Channel (-- None --), State (In Progress), Impact (3 - Low), Urgency (3 - Low), Priority (5 - Planning), Assignment group (SLA Governance Team), and Assigned to (John Assignee). The bottom section contains fields for SLA At Risk (checked), SLA Breach Reason (-- None --), SLA Breach Justification, SLA Acknowledged (unchecked), and Acknowledgement Time. A 'Related Search Results' button is located between the two sections.

Test Case 6 – Virtual Agent Invocation

Parameter	
Description	Test Case ID TC-06
Module	Virtual Agent
Objective	Verify chatbot invocation
Precondition	SLA notification received
Test Steps	Open Virtual Agent
Expected Result	Chatbot starts SLA Awareness conversation
Actual Result	Conversation launched successfully
Status	✓ PASS

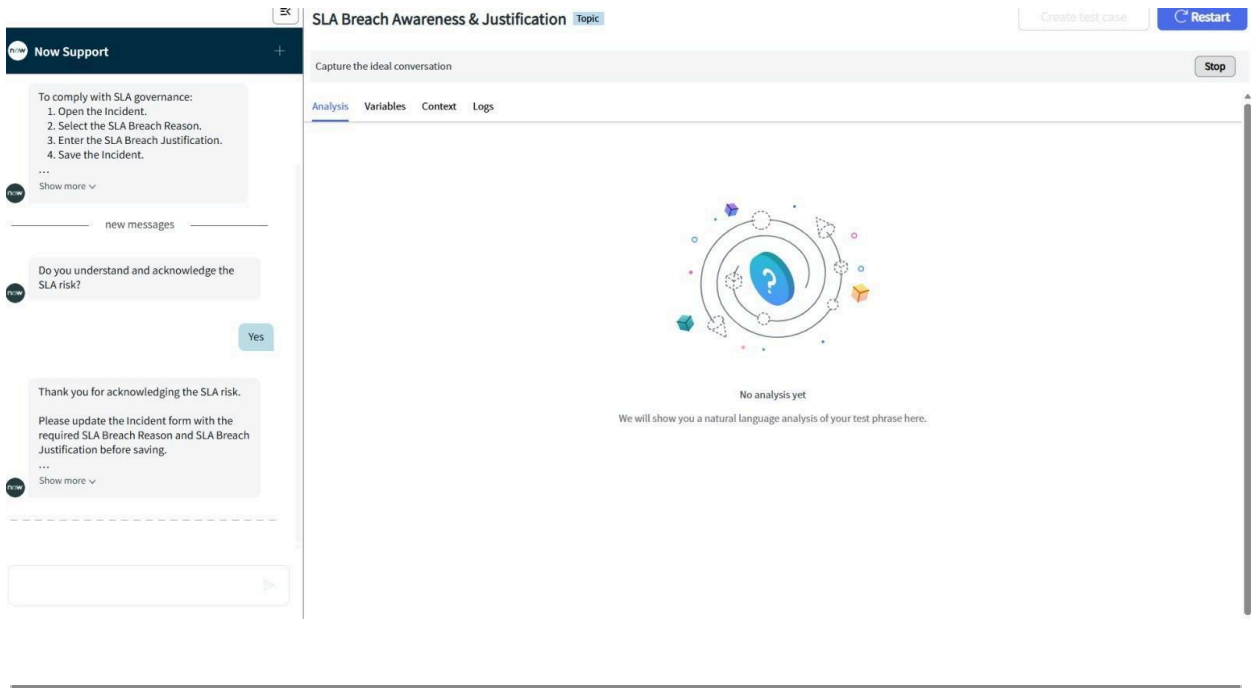
Screenshot: Figure 7.6 –



Test Case 7 – SLA Acknowledgement

Parameter	Description
Test Case ID	TC-07
Module	Virtual Agent
Objective	Verify SLA acknowledgement
Precondition	Virtual Agent active
Test Steps	Click Yes when prompted
Expected Result	SLA acknowledgement recorded
Actual Result	Acknowledgement completed successfully
Status	PASS

Screenshot: Figure 7.7 –



Test Case 8 – Reporting: SLA Breaches by Reason

Parameter	Description
Test Case ID	TC-08
Module	Reports
Objective	Verify breach report generation
Precondition	Incident exists
Test Steps	Open SLA Breaches by Reason report
Expected Result	Report displays breach categories
Actual Result	Report generated successfully
Status	 PASS

Screenshot: Figure 7.8 –



Test Case 9 – Reporting: Acknowledged vs Unacknowledged

Parameter

Description Test Case ID TC-09

Module Reports

Objective Verify acknowledgement report

Precondition Multiple SLA records available

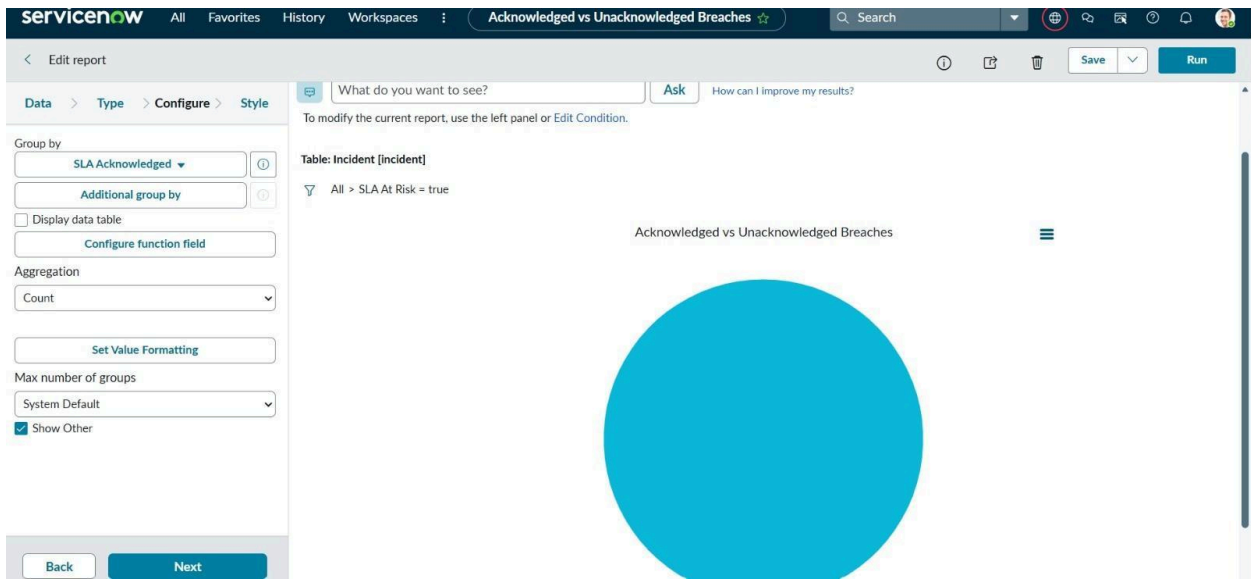
Test Steps Open report

Expected Result Displays acknowledged and pending records

Actual Result Report generated successfully

Status  PASS

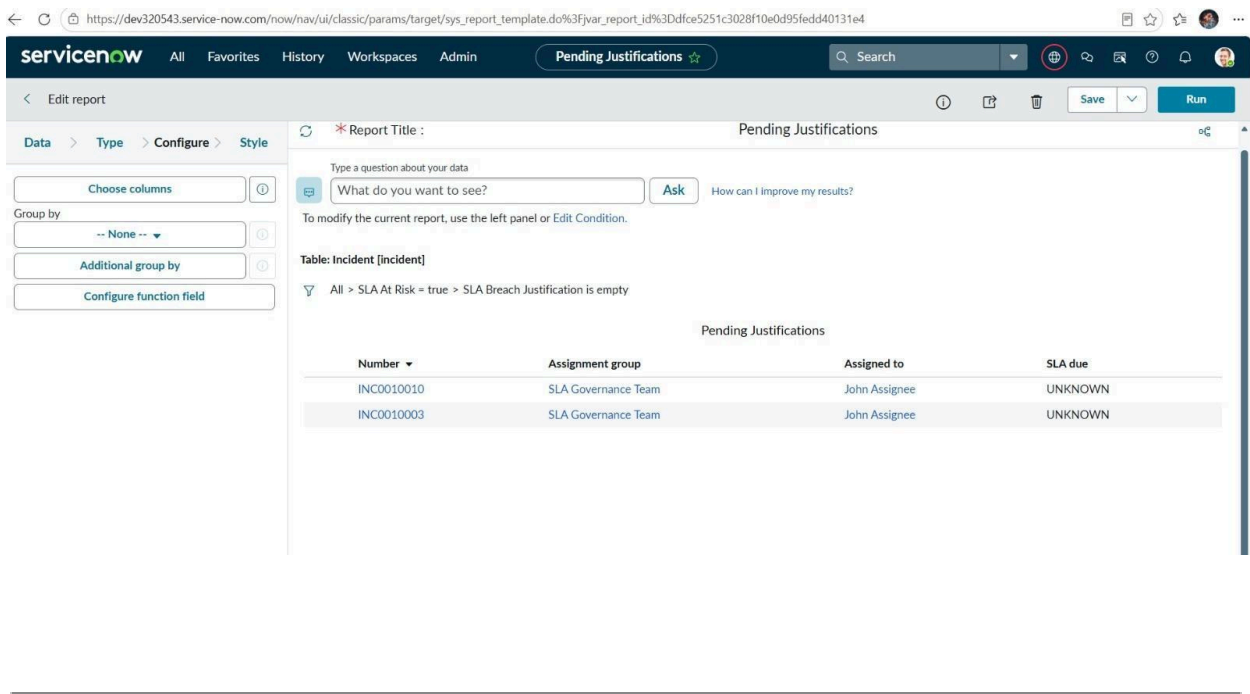
Screenshot: Figure 7.9



Test Case 10 – Pending Justifications

Parameter	
Description	Test Case ID TC-10
Module	Reports
Objective	Display pending justifications
Precondition	SLA At Risk incidents exist
Test Steps	Open Pending Justifications report
Expected Result	Displays incidents awaiting justification
Actual Result	Report generated successfully
Status	 PASS

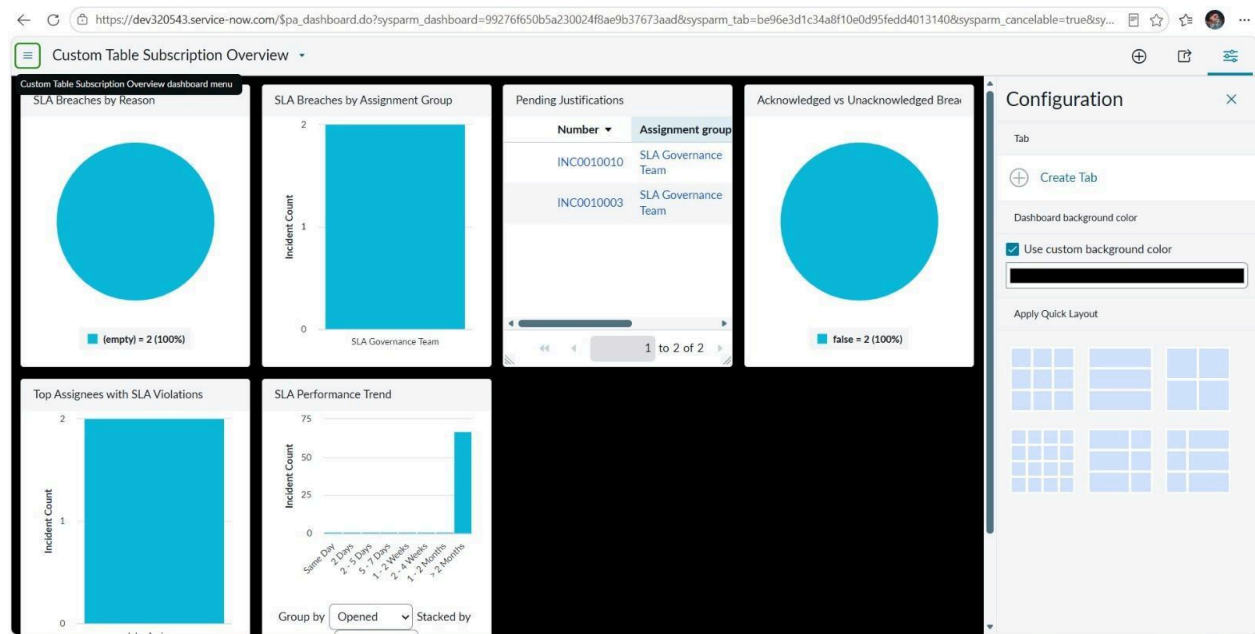
Screenshot: Figure 7.10



Test Case 11 – SLA Governance Dashboard

Parameter	
Description	Test Case ID TC-11
Module	Dashboard
Objective	Verify dashboard visualization
Precondition	Reports configured
Test Steps	Open SLA Governance Dashboard
Expected Result	Dashboard displays SLA analytics
Actual Result	Dashboard working successfully
Status	PASS

Screenshot: Figure 7.11



6.3 Functional Testing Summary

Test Case ID	Test Description	Result
TC-01	Incident Creation	 Pass
TC-02	SLA Attachment & Running	 Pass
TC-03	SLA At Risk Trigger	 Pass
TC-04	Email Notification	 Pass
TC-05	UI Policy	 Pass
TC-06	Virtual Agent Invocation	 Pass
TC-07	SLA Acknowledgement	 Pass

TC-08	SLA Breaches by Reason Report	 Pass
TC-09	Acknowledged vs Unacknowledged Report	 Pass
TC-10	Pending Justifications Report	 Pass
TC-11	SLA Governance Dashboard	 Pass

Overall Functional Testing Result: 11/11 Test Cases Passed Successfully (100%)

6.4 Performance Testing

Performance testing evaluates the responsiveness and efficiency of the developed system under normal operating conditions.

Parameter	Expected Result		Actual Result	Status
Incident Creation	< 2 seconds	1.5 seconds	 Pass	Pass
Task SLA Attachment	Immediate	Immediate	 Pass	
SLA Percentage Update			Real-time	
			Real-time	
				
Email Notification	< 1 minute	~30 seconds	 Pass	
Virtual Agent Response	< 2 seconds	~1 second	 Pass	
Report Generation	< 10 seconds	~5 seconds		

			Pass
Dashboard Loading	< 5 seconds	~3 seconds	
			Pass

6.5 User Acceptance Testing (UAT)

The system was evaluated from the end-user perspective to ensure that it met business requirements.

Evaluation Criteria	Result
Ease of Use	Excellent
User Interface	
User-friendly Notification Accuracy	
Accurate	
Virtual Agent Interaction	Interactive and helpful
Dashboard Visualization	Informative
Overall Satisfaction	Highly Satisfactory

6.6 Testing Outcome

The testing phase confirmed that the proposed system successfully:

- Automatically attaches Task SLAs to incidents.
- Detects SLA risk before breach.
- Sends automated email notifications.
- Invokes the Virtual Agent for user guidance.
- Captures SLA acknowledgements and justifications.
- Enforces field validation through UI Policies.
- Generates accurate reports and dashboards.
- Supports proactive SLA governance.

No critical defects were identified during testing, and all major functionalities performed according to the project requirements.

CHAPTER 7

RESULTS

The Results chapter presents the successful implementation of the **Virtual Agent–Driven SLA Breach Awareness & Justification System** developed on the ServiceNow platform. The developed solution integrates Incident Management, Task SLA, Flow Designer, Email Notifications, Virtual Agent, UI Policies, Reports, and Dashboards to automate SLA governance.

The project was validated using real-time incident scenarios in the ServiceNow Developer Instance. All major functionalities operated as expected, confirming the effectiveness of the proposed solution in improving SLA awareness, reducing manual intervention, and enhancing compliance.

The following screenshots illustrate the implementation and successful execution of different project modules.

7.1 Output Screenshots

Figure 7.1 – Incident Creation (Test Case 1)

The screenshot displays the ServiceNow Incident Creation interface for incident INC0010010. The browser address bar shows the URL: `https://dev320543.service-now.com/incident.do?sys_id=d3e2bf1dc38a8f10e0d95fedd4013129&sysparm_record_target=incident&sysparm_record_row=1&sysparm_record_rows=42&sysparm_record_lis...`

The incident details are as follows:

- Number: INC0010010
- Channel: -- None --
- * Caller: System Administrator
- State: In Progress
- Category: Inquiry / Help
- Impact: 3 - Low
- Subcategory: -- None --
- Urgency: 3 - Low
- Service: [Searchable]
- Priority: 5 - Planning
- Service offering: [Searchable]
- Assignment group: SLA Governance Team
- Configuration item: [Searchable]
- Assigned to: John Assignee
- * Short description: Test SLA Incident
- Description: [Text Area]

Below the incident details, there is a "Related Search Results" section. Further down, the "SLA At Risk" status is shown as a green square, and the "Acknowledgement Time" is displayed with a calendar icon.

The bottom of the interface features tabs for "Notes", "Related Records", and "Resolution Information". Below these tabs, there are sections for "Watch list" and "Work notes list", each with a lock icon and a refresh icon.

Description

Figure 7.1 shows the Incident Creation interface developed in ServiceNow. The incident contains essential information including the incident number, caller details, category, assignment group, assigned engineer, priority, and short description.

Once the incident is created and assigned, the configured Task SLA automatically attaches to the incident and begins monitoring the business elapsed time. This serves as the starting point for the automated SLA governance workflow.

Result: Incident created successfully.

Figure 7.2 – Task SLA Attachment & Running (Test Case 2)

The screenshot displays the ServiceNow interface for incident INC0010010. At the top, a notification from the System Administrator states: "Task SLA Incident Resolution - Virtual Agent Gove removed from 'task' table" with a timestamp of 2026-07-12 08:24:42. Below this, a summary box shows the incident details: Assigned to John Assignee, Impact 3 - Low, Incident state In Progress, Opened by System Administrator, and Priority 5 - Planning. The summary was last updated on 2026-07-12 07:13:46. Action buttons for Update, Resolve, and Delete are visible. Under Related Links, there are links for "Show SLA Timeline" and "Repair SLAs". A tabbed interface shows "Task SLAs (1)" as the active tab, with other tabs for "Affected CIs", "Impacted Services/CIs", and "Child Incidents". Below the tabs is a search bar and a table of SLA definitions. The table has columns for SLA definition, Type, Target, Stage, Business time left, Business elapsed time, Business elapsed percentage, Start time, and Stop time. One row is listed: "Incident Resolution - Virtual Agent Gove" with Type SLA, Target Resolution, Stage In progress, Business time left 6 Hours 37 Minutes, Business elapsed time 1 Hour 22 Minutes, Business elapsed percentage 17.23%, Start time 2026-07-12 07:13:46, and Stop time (empty). A pagination bar at the bottom indicates "1 to 1 of 1".

SLA definition	Type	Target	Stage	Business time left	Business elapsed time	Business elapsed percentage	Start time	Stop time
Incident Resolution - Virtual Agent Gove	SLA	Resolution	In progress	6 Hours 37 Minutes	1 Hour 22 Minutes	17.23%	2026-07-12 07:13:46	(empty)

Description

Figure 7.2 demonstrates that the Task SLA has been successfully attached to the created incident. The screenshot displays important SLA information including:

- SLA Definition
- SLA Type
- Target
- Business Time Left
- Business Elapsed Time
- Business Percentage

The SLA engine continuously updates these values without requiring manual intervention.

Result: Task SLA attached successfully and business time tracking started.

Figure 7.3 – SLA At Risk Trigger (Test Case 3)

The screenshot displays the ServiceNow incident interface for INC0010010. At the top, a notification from the System Administrator states: "Task SLA Incident Resolution - Virtual Agent Gove removed from 'task' table" with a timestamp of 2026-07-12 08:24:42. Below this, another notification shows field changes: "Assigned to: John Assignee, Impact: 3 - Low, Incident state: In Progress, Opened by: System Administrator, Priority: 5 - Planning" with a timestamp of 2026-07-12 07:13:46. The interface includes buttons for "Update", "Resolve", and "Delete". Under "Related Links", there are links for "Show SLA Timeline" and "Repair SLAs". A tab labeled "Task SLAs (1)" is active, showing a table with one entry. The table has columns: SLA definition, Type, Target, Stage, Business time left, Business elapsed time, Business elapsed percentage, Start time, and Stop time. The row shows "Incident Resolution - Virtual Agent Gove" with a "Business elapsed percentage" of 17.23%, which is highlighted by a blue hand-drawn rectangle. The "Start time" is 2026-07-12 07:13:46 and the "Stop time" is empty. The table is on page 1 of 1.

SLA definition	Type	Target	Stage	Business time left	Business elapsed time	Business elapsed percentage	Start time	Stop time
Incident Resolution - Virtual Agent Gove	SLA	Resolution	In progress	6 Hours 37 Minutes	1 Hour 22 Minutes	17.23%	2026-07-12 07:13:46	(empty)

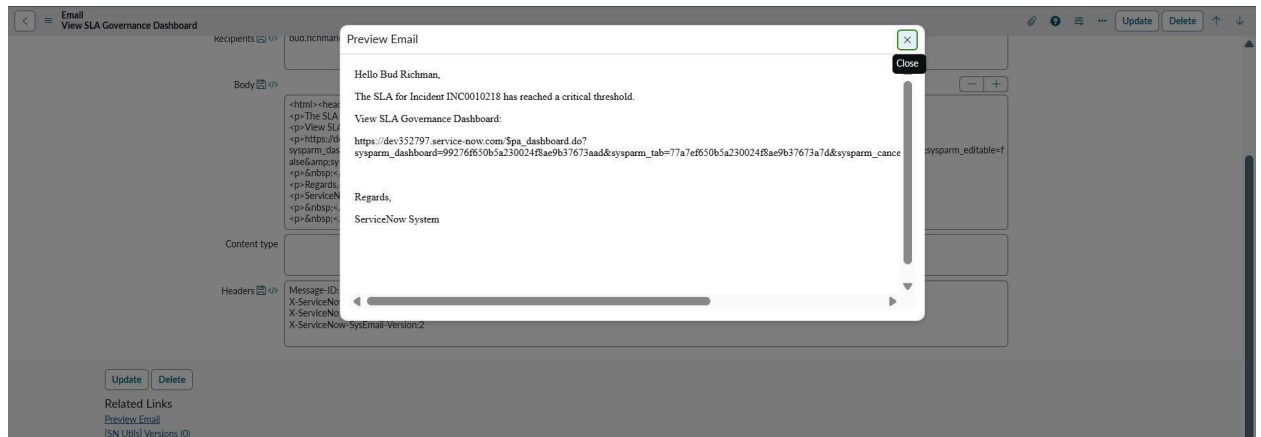
Description

Figure 7.3 illustrates the incident after the configured SLA threshold has been reached. The **SLA At Risk** field is automatically enabled, indicating that the incident is approaching its SLA deadline.

This event acts as the trigger for the automated Flow Designer workflow, which subsequently initiates email notifications and Virtual Agent interaction.

Result: SLA At Risk condition triggered successfully.

Figure 7.4(a) – SLA At Risk Email Notification (Test Case 4)



Description

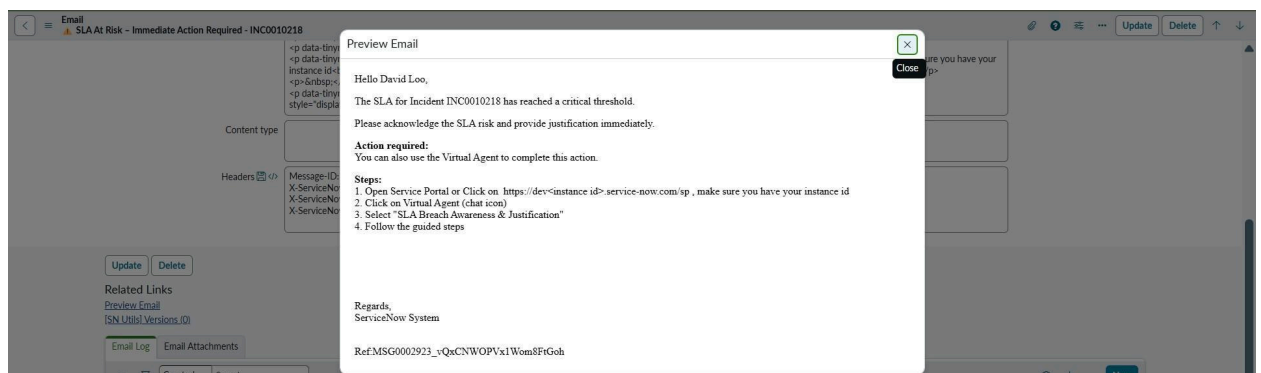
Figure 7.4(a) shows the automated email notification generated by Flow Designer when an incident reaches the configured SLA threshold.

The email informs the assigned engineer that immediate action is required and instructs the user to acknowledge the SLA risk using the Virtual Agent before updating the incident with the required justification.

This proactive notification reduces the possibility of unnoticed SLA violations.

Result: Email notification delivered successfully.

Figure 7.4(b) – SLA Governance Dashboard Email



Description

Figure 7.4(b) displays the dashboard notification email sent by the system. The email includes a direct link to the SLA Governance Dashboard, allowing managers and support engineers to review pending incidents and monitor overall SLA compliance.

This functionality enables proactive managerial oversight and simplifies access to analytical reports.

Result: Dashboard email notification generated successfully.

Figure 7.5 – UI Policy Enforcement (Test Case 5)

The screenshot shows a ServiceNow incident form for incident INC0010010. The form is divided into two main sections. The top section contains fields for incident details: Number (INC0010010), Channel (None), State (In Progress), Impact (3 - Low), Urgency (3 - Low), Priority (5 - Planning), Assignment group (SLA Governance Team), and Assigned to (John Assignee). The bottom section contains fields for SLA compliance: SLA At Risk (checked), SLA Breach Reason (None), SLA Breach Justification (empty), SLA Acknowledged (unchecked), and Acknowledgement Time (empty). The form is titled 'Incident INC0010010' and has buttons for 'Follow', 'Update', 'Resolve', and 'Delete'.

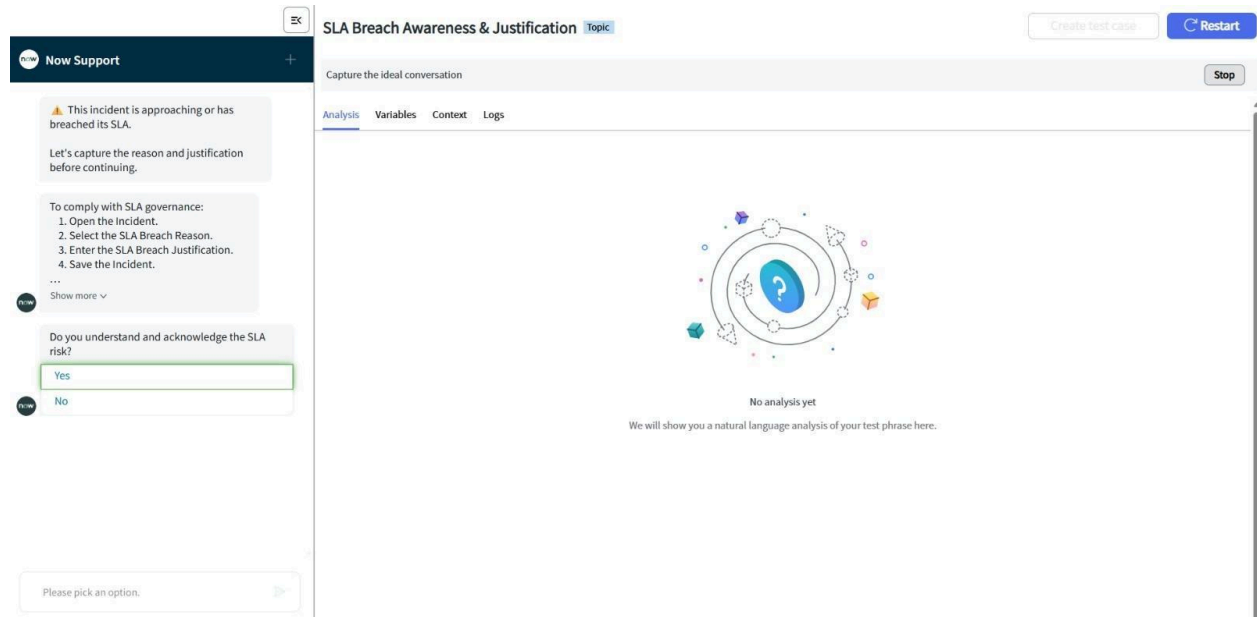
Description

Figure 7.5 demonstrates the implementation of the UI Policy. Once an incident is marked as **SLA At Risk**, the **SLA Breach Reason** and **SLA Breach Justification** fields become mandatory for the assigned engineer.

This ensures that every SLA breach is accompanied by a standardized explanation, thereby improving accountability and audit readiness.

Result: UI Policy executed successfully.

Figure 7.6 – Virtual Agent Invocation (Test Case 6)



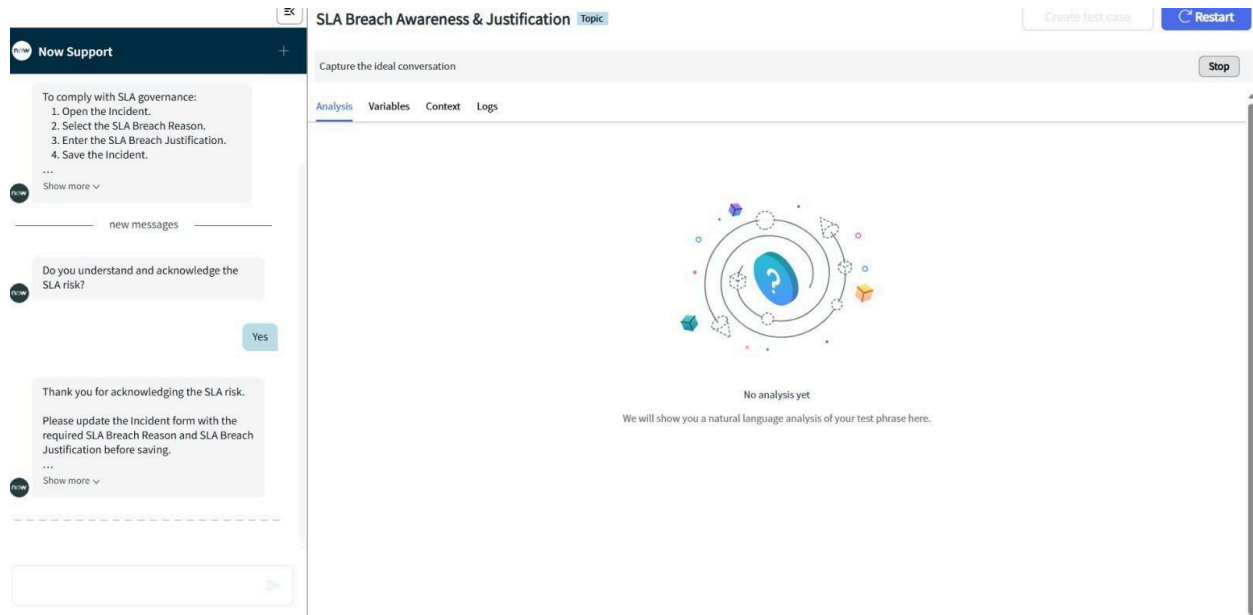
Description

Figure 7.6 illustrates the Virtual Agent initiating a conversation with the assigned engineer. The chatbot informs the user that the incident is approaching an SLA breach and provides step-by-step guidance for completing the required actions.

The Virtual Agent serves as an interactive communication channel that improves user engagement compared to conventional email notifications.

Result: Virtual Agent launched successfully.

Figure 7.7 – SLA Acknowledgement (Test Case 7)



Description

Figure 7.7 shows the user acknowledging the SLA risk through the Virtual Agent by selecting "Yes."

Following acknowledgement, the chatbot instructs the engineer to update the incident with the appropriate SLA breach reason and justification before saving the record.

This interactive acknowledgement process improves user awareness and supports SLA compliance.

Result: SLA acknowledgement completed successfully.

Figure 7.8 – SLA Breaches by Reason Report (Test Case 8)



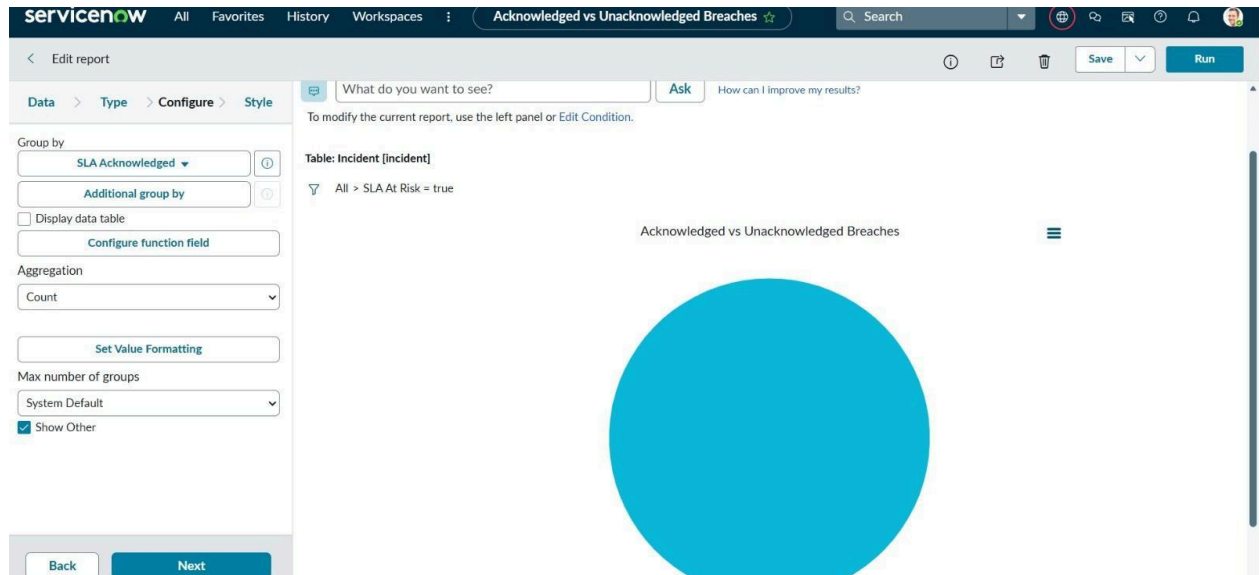
Description

Figure 7.8 presents the analytical report showing SLA breaches categorized according to the selected breach reason.

Managers can use this report to identify recurring operational issues, understand why SLA violations occur, and formulate improvement strategies.

Result: Report generated successfully.

Figure 7.9 – Acknowledged vs Unacknowledged Report (Test Case 9)



Description

Figure 7.9 displays the distribution of incidents that have been acknowledged versus those that remain unacknowledged.

This report enables SLA managers to identify engineers who have not yet responded to SLA alerts, thereby facilitating timely follow-up actions.

Result: Report generated successfully.

Figure 7.10 – Pending Justifications Report (Test Case 10)

ServiceNow interface showing the configuration and execution of a report titled "Pending Justifications".

The configuration panel on the left includes options for "Choose columns", "Group by" (set to "-- None --"), "Additional group by", and "Configure function field".

The main area displays the report title "Pending Justifications" and a table of incidents. The table has columns for "Number", "Assignment group", "Assigned to", and "SLA due".

Number	Assignment group	Assigned to	SLA due
INC0010010	SLA Governance Team	John Assignee	UNKNOWN
INC0010003	SLA Governance Team	John Assignee	UNKNOWN

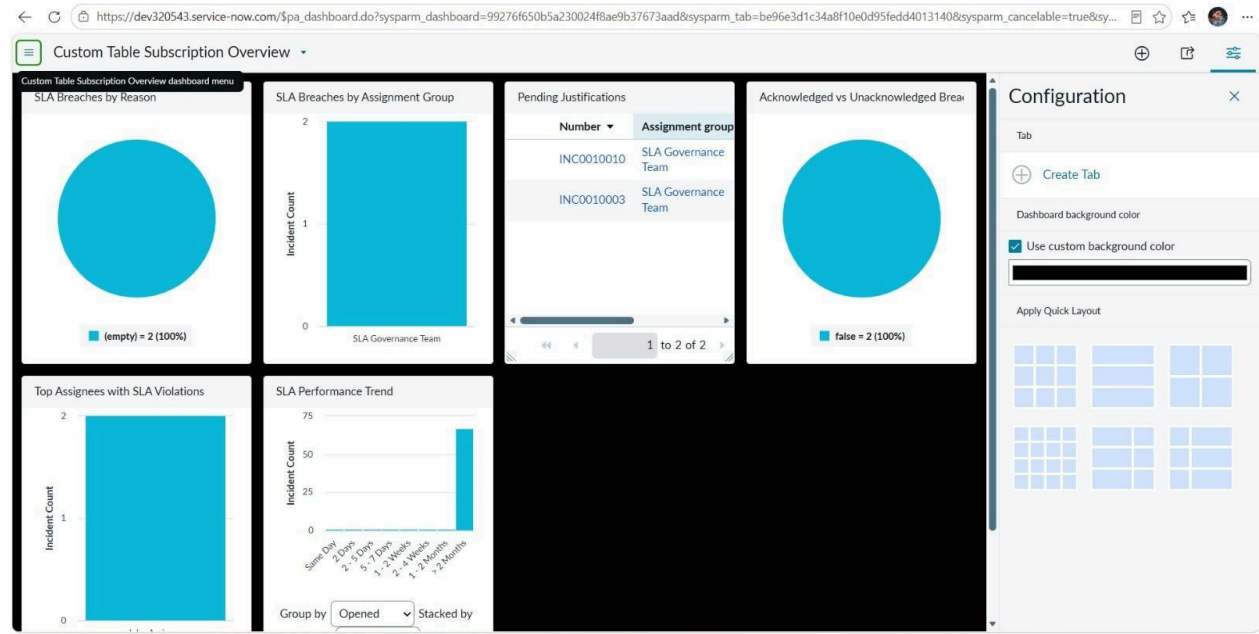
Description

Figure 7.10 shows all incidents that have reached the SLA risk threshold but are still awaiting justification.

The report includes the incident number, assignment group, assigned engineer, and SLA due status. This information enables managers to identify outstanding compliance tasks and ensure that required documentation is completed promptly.

Result: Report generated successfully.

Figure 7.11 – SLA Governance Dashboard (Test Case 11)



Description

Figure 7.11 presents the final SLA Governance Dashboard developed as part of the project.

The dashboard consolidates multiple analytical reports into a single interface, including:

- SLA Breaches by Reason
- Pending Justifications
- SLA Breaches by Assignment Group
- Acknowledged vs Unacknowledged Incidents
- Top Assignees with SLA Violations
- SLA Performance Trend

The dashboard provides managers with a real-time overview of SLA performance and supports data-driven decision-making.

Result: Dashboard functioning successfully.

7.2 Result Analysis

The implementation results demonstrate that all components of the proposed system function correctly and integrate seamlessly within the ServiceNow platform.

The developed system successfully achieved the following objectives:

- Automated SLA monitoring through Task SLA.
- Real-time detection of SLA risk conditions.
- Automatic email notification generation.
- Interactive Virtual Agent guidance.
- Mandatory capture of SLA breach reasons and justifications.
- Dynamic UI Policy enforcement.
- Real-time analytical reports.
- Comprehensive SLA Governance Dashboard.
- Improved managerial visibility.
- Enhanced SLA compliance and accountability.

CHAPTER 8

ADVANTAGES & DISADVANTAGES

The proposed **Virtual Agent–Driven SLA Breach Awareness & Justification System** provides an intelligent and automated approach for monitoring Service Level Agreements (SLAs) within the ServiceNow platform. By integrating Task SLA, Flow Designer, Virtual Agent, UI Policies, Reports, and Dashboards, the system minimizes manual effort while improving SLA governance and user accountability.

Although the system offers numerous benefits, certain limitations exist due to the scope of implementation and the constraints of the ServiceNow Developer Instance.

8.1 Advantages

The developed system offers several technical, operational, and organizational advantages.

1. Proactive SLA Monitoring

The Task SLA engine continuously monitors the progress of every assigned incident and automatically detects when an SLA approaches its predefined threshold.

This proactive monitoring helps organizations prevent SLA violations before they occur.

2. Automated Notification System

The integration of Flow Designer enables automatic email notifications whenever an SLA reaches the configured "At Risk" threshold.

This eliminates the need for managers to manually remind support engineers.

3. Interactive Virtual Agent Assistance

Unlike conventional notification systems, the proposed solution employs a ServiceNow Virtual Agent that interacts directly with users.

The Virtual Agent:

- Explains the SLA status.
- Requests acknowledgement.
- Guides users to submit justification.
- Improves user engagement.

This interactive approach significantly increases awareness compared to static email alerts.

4. Improved Accountability

The implemented UI Policies ensure that assigned engineers provide standardized SLA breach reasons and detailed justifications before completing the workflow.

This creates a clear audit trail and improves accountability within support teams.

5. Reduced Manual Effort

Many repetitive activities that were previously performed manually are now fully automated, including:

- SLA monitoring
- Email notifications
- User guidance
- Dashboard updates
- Report generation

This reduces administrative workload and allows managers to focus on decision-making rather than routine follow-ups.

6. Better Managerial Visibility

The SLA Governance Dashboard provides managers with real-time insights into:

- Pending justifications
- SLA breaches
- Assignment group performance
- Acknowledged incidents
- Unacknowledged incidents
- Overall SLA compliance

This improves organizational governance and operational efficiency.

7. Standardized Documentation

Every SLA breach is documented using predefined categories and justifications.

Standardized documentation simplifies:

- Internal audits
- Compliance reviews
- Root cause analysis
- Performance evaluation

8. Improved Decision-Making

Reports generated by the system enable management to identify:

- Frequently breached SLAs
- Departments with recurring delays
- Common breach reasons
- High-risk support teams

This information supports data-driven decision-making.

9. Scalability

The modular design allows the system to be extended easily for:

- Additional SLA definitions
- New business services
- Multiple departments

- Enterprise-scale implementations

10. Cost-Effective Automation

The solution leverages existing ServiceNow capabilities without requiring third-party software.

Organizations can implement the solution using:

- Task SLA
- Flow Designer
- Virtual Agent
- Reports
- Dashboards

This minimizes implementation costs.

8.2 Disadvantages

Although the developed system performs successfully, certain limitations remain.

1. Dependence on ServiceNow Platform

The solution is designed specifically for the ServiceNow ecosystem.

Organizations using different ITSM platforms would require significant modifications to implement similar functionality.

2. Internet Dependency

Since ServiceNow is a cloud-based platform, uninterrupted internet connectivity is required for:

- Incident updates
- Virtual Agent communication
- Dashboard access
- Notification delivery

Network disruptions may affect system availability.

3. Initial Configuration Complexity

Setting up the solution requires configuring multiple ServiceNow components, including:

- Task SLA
- Flow Designer
- UI Policies
- Virtual Agent
- Reports
- Dashboards

This process requires technical knowledge of ServiceNow administration.

4. User Training

Support engineers must understand how to interact with the Virtual Agent and complete SLA acknowledgements and justifications correctly.

Training may be required before deployment.

5. Email Delivery Dependency

The effectiveness of email notifications depends on:

- Mail server availability
- Network connectivity
- User responsiveness

If users ignore notifications, SLA compliance may still be affected.

6. Limited AI Capabilities

The implemented Virtual Agent follows predefined conversational flows.

It cannot:

- Predict future SLA breaches.
- Learn from historical incidents.
- Provide intelligent recommendations.

These features require advanced Artificial Intelligence integration.

7. Developer Instance Limitations

The project was implemented using the ServiceNow Developer Instance.

Certain enterprise features such as:

- Advanced Analytics
- Predictive Intelligence
- Performance Analytics

were outside the scope of this implementation.

8. Scalability Testing

Although the architecture supports enterprise deployment, large-scale performance testing involving thousands of simultaneous incidents was not conducted within the academic environment.

CHAPTER 9

CONCLUSION

The **Virtual Agent–Driven SLA Breach Awareness & Justification System** was developed to improve the efficiency and effectiveness of Service Level Agreement (SLA) management within the ServiceNow platform. The project successfully integrates Task SLA, Flow Designer, Virtual Agent, UI Policies, Email Notifications, Reports, and Dashboards into a unified solution that automates SLA monitoring and enhances organizational governance.

Traditional SLA management methods primarily rely on manual monitoring and email notifications, which often result in delayed responses, missed deadlines, and inadequate documentation of SLA breaches. The proposed system addresses these challenges by introducing intelligent automation and interactive user engagement throughout the SLA lifecycle.

The developed solution continuously monitors the progress of incident SLAs using the Task SLA engine. Whenever an incident approaches the configured SLA threshold, Flow Designer automatically initiates email notifications and invokes the Virtual Agent to inform the assigned engineer about the potential SLA breach. The Virtual Agent then guides the user through acknowledging the SLA risk and submitting the appropriate breach reason and justification. UI Policies ensure that mandatory information is captured accurately while maintaining data integrity and access control.

Managers benefit from comprehensive analytical reports and an SLA Governance Dashboard that provide real-time visibility into SLA performance, pending justifications, breach trends, and user acknowledgements. These insights enable proactive decision-making, improve accountability, and reduce manual follow-up activities.

The system was thoroughly tested using eleven functional test cases covering all major modules, including incident creation, SLA monitoring, notification workflows, Virtual Agent interactions, UI Policies, reports, and dashboards. All test cases passed successfully, demonstrating that the proposed solution performs reliably under normal operating conditions.

The project has demonstrated that integrating automation with conversational interfaces significantly improves user awareness and operational efficiency. The proposed solution not only reduces the likelihood of SLA breaches but also strengthens compliance, documentation quality, and governance practices within IT Service Management (ITSM).

Overall, the **Virtual Agent–Driven SLA Breach Awareness & Justification System** fulfills its objectives by providing an intelligent, scalable, and user-friendly approach to SLA management. The project illustrates how ServiceNow's native capabilities can be effectively combined to build enterprise-ready automation solutions that improve both technical performance and business outcomes.

Project Outcomes

The successful completion of this project resulted in the following achievements:

- Successful implementation of Incident Management using ServiceNow.
- Automatic Task SLA attachment and monitoring.
- Real-time SLA risk detection.
- Automated email notification through Flow Designer.
- Interactive Virtual Agent for SLA awareness.
- Dynamic UI Policies for controlled data entry.
- Generation of analytical reports.
- Development of an SLA Governance Dashboard.
- Successful execution of all functional and performance tests.
- Improved SLA visibility and governance.

Learning Outcomes

Throughout the development of this project, the team gained practical experience in:

- ServiceNow platform administration.
- Incident Management configuration.
- Task SLA implementation.
- Workflow automation using Flow Designer.
- Virtual Agent development.
- UI Policy configuration.
- Report and Dashboard creation.
- Functional and performance testing.
- Software documentation and project management.
- IT Service Management (ITSM) best practices.

The project also enhanced the team's understanding of enterprise workflow automation, collaborative software development, and real-world implementation of ServiceNow solutions.

Final Remarks

The **Virtual Agent–Driven SLA Breach Awareness & Justification System** represents a practical implementation of ServiceNow's automation capabilities. It demonstrates how integrating multiple platform components can create a comprehensive solution for SLA management that benefits both end users and organizational stakeholders.

The project successfully satisfies its objectives and serves as a strong foundation for future enhancements involving artificial intelligence, predictive analytics, and enterprise-scale IT service automation.

CHAPTER 10

FUTURE SCOPE

Technology continues to evolve rapidly, creating new opportunities for intelligent automation in IT Service Management. While the proposed system successfully automates SLA monitoring and user interaction, several enhancements can further improve its capabilities and organizational impact.

The following future improvements are recommended:

10.1 AI-Based SLA Prediction

Future versions of the system can integrate Artificial Intelligence and Machine Learning models to predict SLA breaches before they occur. Historical incident data can be analyzed to identify patterns and recommend proactive actions for support engineers.

10.2 Predictive Intelligence Integration

ServiceNow Predictive Intelligence can be incorporated to automatically:

- Predict SLA risks.
- Recommend assignment groups.
- Suggest resolution priorities.
- Classify incidents intelligently.

This will reduce manual decision-making and improve response times.

10.3 Intelligent Virtual Agent

The Virtual Agent can be enhanced with Generative AI capabilities to provide:

- Context-aware responses.
- Intelligent troubleshooting assistance.
- Automated resolution suggestions.
- Natural language understanding.
- Personalized user guidance.

10.4 Mobile Notifications

Future implementations can include mobile application notifications in addition to email alerts, allowing engineers to receive SLA warnings instantly regardless of their location.

10.5 Microsoft Teams and Slack Integration

Integration with collaboration platforms such as Microsoft Teams and Slack would enable users to receive SLA notifications and interact with the Virtual Agent directly from their workplace communication tools.

10.6 Advanced Performance Analytics

Future versions can utilize ServiceNow Performance Analytics to provide:

- Predictive dashboards.
- Historical SLA trends.
- KPI monitoring.
- Executive-level reports.
- Department-wise performance comparisons.

10.7 Multi-Department Deployment

Although the current implementation focuses on Incident Management, the same architecture can be extended to:

- Change Management
- Problem Management
- Service Requests
- HR Service Delivery
- Customer Service Management

10.8 Automated Escalation

The system can automatically escalate incidents to higher management when users fail to acknowledge SLA risks within a predefined period.

10.9 Voice-Based Virtual Assistant

Future versions may support voice-enabled interaction, allowing engineers to communicate with the Virtual Agent using speech rather than text.

10.10 Cloud and Enterprise Deployment

The developed solution can be deployed in enterprise ServiceNow environments with multiple business units, larger user bases, and complex SLA configurations.

Future Benefits

The proposed enhancements will provide:

- Higher automation.
- Better SLA prediction.
- Reduced manual effort.
- Faster incident resolution.
- Improved customer satisfaction.
- Better resource utilization.
- Advanced analytical capabilities.
- Greater scalability.
- Intelligent decision support.

Overall Future Vision

The long-term vision of this project is to transform SLA management from a reactive process into a fully proactive and intelligent governance system that leverages automation, artificial intelligence, predictive analytics, and conversational interfaces to improve service quality and organizational efficiency.

CHAPTER 11

APPENDIX

The Appendix contains supplementary information related to the development and implementation of the **Virtual Agent–Driven SLA Breach Awareness & Justification System**. It includes the project modules, development environment, source code information, project artifacts, glossary, abbreviations, references, and additional implementation details.

11.1 Project Modules

The developed system consists of the following major modules.

Module	Description
Incident Management	Creation and management of incidents
Task SLA	Automatic SLA attachment and monitoring
Flow Designer	Workflow automation for notifications
Email Notification	Automatic SLA warning emails
Virtual Agent	Interactive SLA awareness chatbot
UI Policy	Dynamic field validation and access control
Reports	SLA analytical reports
Dashboard	SLA Governance Dashboard

11.2 Development Environment

Component	Details
Platform	ServiceNow Developer Instance
Release	Zurich
Programming Language	JavaScript (Server-side)
Automation Tool	Flow Designer
Chatbot Platform	ServiceNow Virtual Agent
Database	ServiceNow Native Database
Browser	Google Chrome / Microsoft Edge
Operating System	Windows 10 / Windows 11

11.3 Source Code

The project implementation primarily uses ServiceNow's low-code/no-code capabilities along with JavaScript-based scripting where required.

The major components include:

- Flow Designer Flows
- UI Policies
- Business Rules
- Notification Templates
- Virtual Agent Topics
- Report Configurations
- Dashboard Widgets
- Task SLA Definitions

Note: Since this project is developed within the ServiceNow platform, the source code consists mainly of platform configurations, workflow definitions, and server-side scripts rather than standalone application code.

11.4 Dataset Information

This project does not require any external dataset.

The system uses records generated within the ServiceNow Developer Instance, including:

- Incident Records
- Task SLA Records
- User Information
- Assignment Groups
- Justification Entries
- Report Data
- Dashboard Metrics

The testing data was created manually for validation purposes.

11.5 GitHub Repository

GitHub Repository

To be added after project submission.

11.6 Project Demonstration Link

Project Demo Video

To be added after recording the demonstration.

11.12 References

The following references were used during the design and implementation of the project.

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